

BANK MANAGEMENT SYSTEM

**A Mini Project Report Submitted in Partial Fulfillment
of the Requirements for the Degree
of
MASTER OF COMPUTER APPLICATION**

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SESSION: 2024-2025

DECLARATION

I hereby declare that this submission of our own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma or the university or other institute of higher learning, except where due acknowledgement has been made in the text.

Signature:

Name: KIRTI

Roll no.:

Date:

CERTIFICATE

*This is to certify that the Project Report entitled “**Bank Management System**” which is submitted by **KIRTI (2301650140039)** of 3rd semester, in the year 2024-2025.*

*In partial fulfillment of the requirement for the award of degree of MCA in department of **Computer Application of Kanpur Institute of Technology (Dr. APJ Abdul Kalam Technical University)** is a record of the candidate own work carried out by him/her under my supervision. The matter embodied in this thesis is original and is not been submitted for the award of any degree.*

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*It gives me a great sense of pleasure to present the report of the MCA Project undertaken during **MCA Second Year**. I owe special debt of gratitude to our guide **Mr. Amit Kumar Tiwari** of Department of **Computer Application, Kanpur Institute of Technology, Kanpur (U.P)** for his constant support and guidance throughout course of our work. His sincerity, thoroughness and perseverance have been a constant source of inspiration for me. It is only his cognizant efforts that our endeavors have been light of the day.*

*I also take the opportunity to acknowledge the contribution of **Mr. Ajeet Singh , Head of Department of Computer Application, Kanpur Institute of Technology, Kanpur (U.P.)** for his full support and assistance during the development of the project.*

I also do not like to miss the opportunity to acknowledge the contribution of all faculty members of the department for their assistance and cooperation during the development of my project.

Signature:

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Roll no:

Date:

ABSTRACT

This thesis presents a JAVA Program which helps to manage the bank operation, including card number and pin generator, transaction processing and ATM facility.. This provide a JAVA based solution . It's a GUI-based project used with the Swing module to organize all the elements that work under bank management.

Although financial technology is reducing the need to make regular cash transactions, it's also making it increasingly convenient to get cash when you need it.

Cardless ATMs allow you to perform ATM transactions via your smartphone's bank app. They're a convenient way to get money from your bank account if you forget your debit card or if you prefer to minimize the number of times you need to touch an ATM's screen.

By the use of JAVA core and My SQL providing a more secure way of transaction with no need to carry a card or wallet and providing a more sanitary option through this project. To get a more practical approach towards using JAVA and My SQL together from this project.

It is a JAVA program for solving financial applications of a customer in banking environment in order to nurture the needs of an end banking user by providing various ways to perform banking tasks. Also to enable the users, workspace to have additional functionalities which are not provided under a conventional banking software.

Cardless ATM withdrawals won't eliminate the need to touch the ATM, yet they do cut down on contact. And the fewer touches the better, considering how many users might touch an ATM and spread germs between cleanings.

Cardless withdrawals also often require two-step authentication, so hackers wouldn't only need to know your PIN, but they would also need to have access to your phone.

It isn't always convenient to carry a wallet, purse or something else containing your cards. Cardless ATMs allow you to perform transactions if you have your smartphone

Overall, this can work as a base for the various Banks and other government agencies to provide easy bank operations and a cardless ATM service.

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BANK MANAGEMENT SYSTEM

CHAPTER 1

1.INTRODUCTION

1.1General Introduction

This is a Program on Bank Management System offers the comprehensive java based solution for managing bank operations, including card number and pin generator, transaction processing and ATM facility. It is a GUI-based project used with the swing module to organize all the elements that work under bank management. Although financial technology is reducing the need to make regular cash transactions, it's also making it increasingly convenient to get cash when you need it. Cardless ATMs allow you to perform ATM transactions via your smartphone's bank app. They're a convenient way to get money from your bank account if you forget your debit card or if you prefer to minimize the number of times you need to touch an ATM's screen.

1.2 Project Overview

This project has a java based programming with several classes named as login , sign up, wit etc. This project contain several important modules as shown below.

Modules of the project:

- Module 1: Login = contain insertion for card number and pin with three options of sign in, sing up and clear.
- Module 2: Signup = contain page 1 of personal information form.
- Module 3: Signup2 = contain page 2 of additional information form.
- Module 4: Signup3 = contain page 3 of account details form.
- Module 5: generation of card number and pin after successful submit of forms.
- Module 6: access to ATM facility.
- Module 7: Transaction facilities: deposit, withdrawal of money, balance enquiry, fast cash, pin change and mini statement.

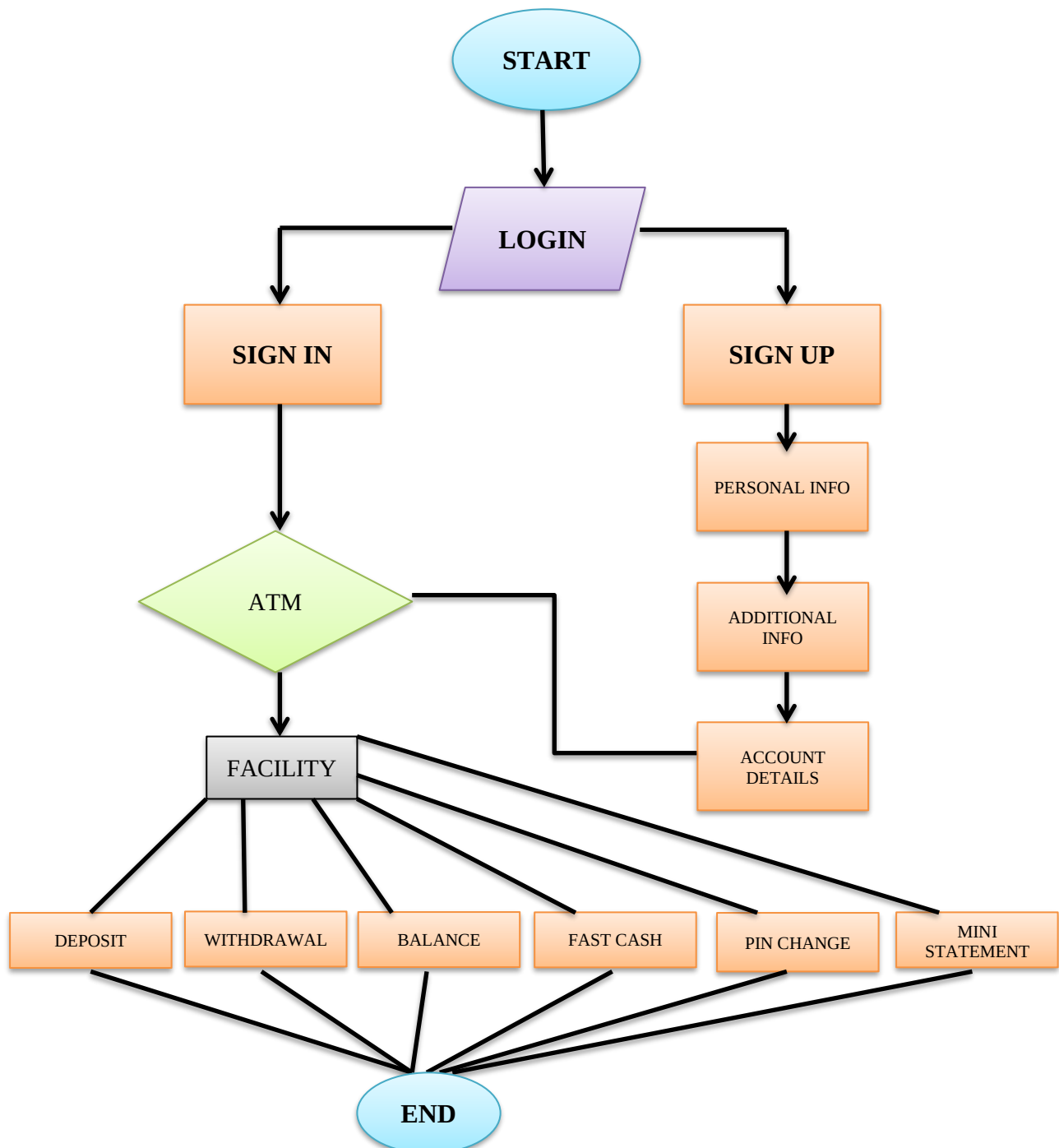
1.3 Aim of this Project

To make a java based solution for managing bank operations, including card number and pin generator, transaction processing and ATM facility. By the use of JAVA core and My SQL providing a more secure way of transaction with no need to carry a card or wallet and providing a more sanitary option through this project. To get a more practical approach towards using JAVA and My SQL together.

1.4 Methodology(Summary)

Firstly user will go to Login page through which the user can get access for further page through either Sign in or Sign up button. The authorised user can get access through the Sign in button. Other users need to make an account through Sign up button , where they have to fill three page of form named as personal information, additional information and account details. After this the users will get a card number and pin through a pop up message, and get access to further facilities. Like deposition of money , withdrawal of money, pin change, balance enquiry, mini statement and fast cash.

Flow Chart



1.5 WHAT IS CARDLESS ATM ?

A cardless ATM is a machine that allows users to withdraw cash from their account without using a physical card. Instead, users can use a banking app or text message to verify their account.

Cardless ATM withdrawals won't eliminate the need to touch the ATM, yet they do cut down on contact. And the fewer touches the better, considering how many users might touch an ATM and spread germs between cleanings. Cardless withdrawals also often require two-step authentication, so hackers wouldn't only need to know your PIN, but they would also need to have access to your phone. It isn't always convenient to carry a wallet, purse or something else containing your cards. Cardless ATMs allow you to perform transactions if you have your smartphone.

What are cardless ATMs and how do they work?

Cardless ATMs provide access to your account and allow you to withdraw cash without the need for a card. Instead, they rely on account verification via text message or a banking app on your smartphone.

There are several ways that cardless ATMs can function. Types of cardless technology include quick response (QR) codes, near-field communication (NFC), verification codes and biometric verification.

QR Codes

To use a cardless ATM with a QR code you first set the mobile withdrawal up on your banking app. The ATM will then display a QR code on the screen for you to scan with your phone before dispensing the money.

Near-field Communication (NFC)

Some services like Apple Pay use NFC, or near-field communication, technology. To use NFC at a cardless ATM, open the app on your mobile device, choose the linked bank account you want to withdraw from and tap your phone against the designated reader. To complete the transaction, the ATM will prompt you to type in your PIN (just as if you had inserted your card).

Verification codes

Using a verification code involves opening the bank's app on your phone and selecting the type of ATM transaction you'd like to perform. The app then provides a one-time code you'll enter at the ATM as well as your PIN. The code usually expires after a set amount of time, such as 30 minutes.

Once you enter the verification code, your transaction will take place, which can mean the ATM will dispense the amount of money you indicated you wish to withdraw.

Biometric verification

You may use biometric verification to unlock your smartphone by having it recognize your face or fingerprint. Similarly, some banks can store your biometric data so their ATMs can use them to verify your identity for future transactions. This allows you to withdraw cash or perform other functions without having to provide your card.

1.6 Scope of the project

1. The main objective of the project is to develop an online Banking system for banks. This bank management system will help us- to add new customer accounts, and deposit amount, withdraw amount and pin change like facilities. Using this system, users can also search any individual account in a few seconds.
2. To develop a software for solving financial applications of a customer in banking environment in order to nurture the needs of an end banking user by providing various ways to perform banking tasks. Also to enable the users, workspace to have additional functionalities which are not provided under a conventional banking software.
3. Cardless ATM withdrawals won't eliminate the need to touch the ATM, yet they do cut down on contact. And the fewer touches the better, considering how many users might touch an ATM and spread germs between cleanings.
4. Cardless withdrawals also often require two-step authentication, so hackers wouldn't only need to know your PIN, but they would also need to have access to your phone.
5. It isn't always convenient to carry a wallet, purse or something else containing your cards. Cardless ATMs allow you to perform transactions if you have your smartphone.

CHAPTER 2

2. TECHNOLOGY USED

- Front end = JAVA JDK_23
- Programming language = JAVA
- Back end = My SQL

2.1 JAVA

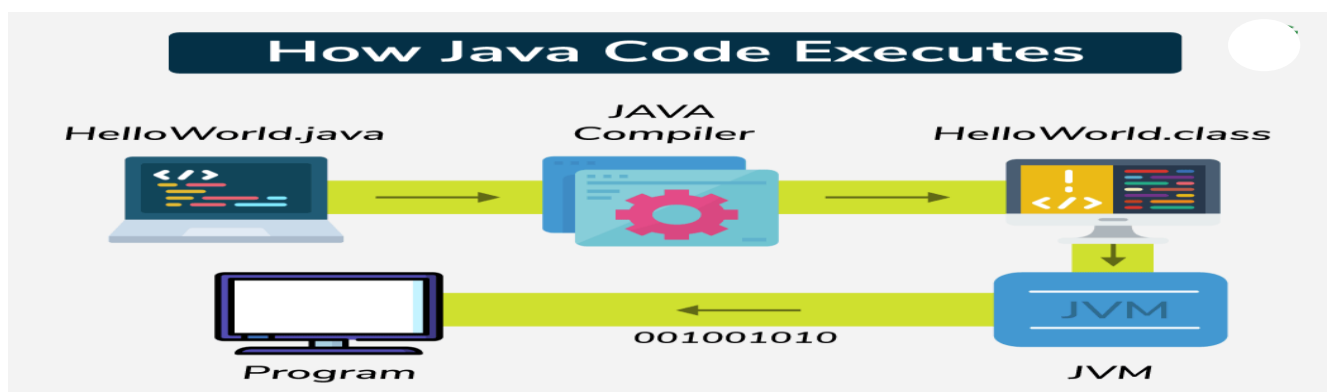
Java is a class-based, **object-oriented programming language** that is designed to have as few implementation dependencies as possible. It is intended to let application developers **Write Once and Run Anywhere (WORA)**, meaning that compiled Java code can run on all platforms that support Java without the need for recompilation. Java was developed by **James Gosling** at **Sun Microsystems Inc.** in **May 1995** and **later acquired by Oracle Corporation** and is widely used for developing applications for desktop, web, and mobile devices.

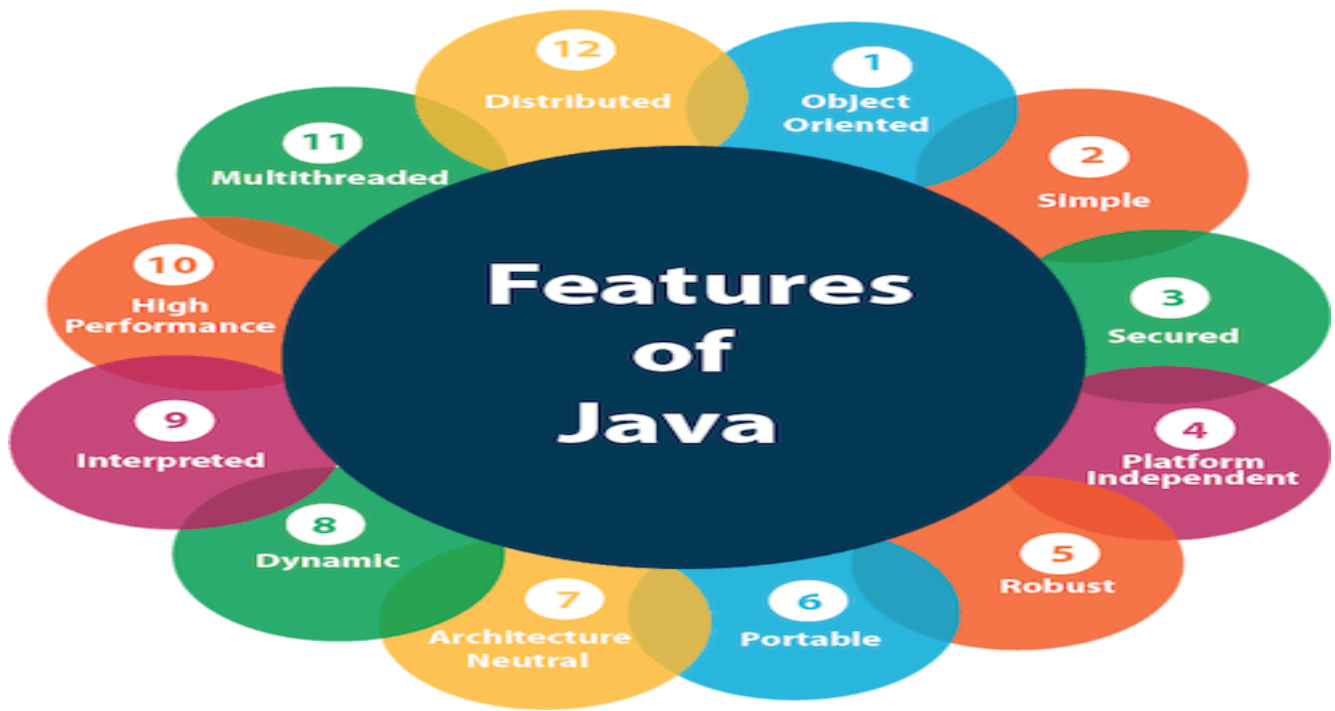
Java is known for its simplicity, robustness, and security features, making it a popular choice for enterprise-level applications. Java applications are compiled to byte code that can run on any Java Virtual Machine. The syntax of Java is similar to C/C++.

Java makes writing, compiling, and debugging programming easy. It helps to create reusable code and modular programs.

How Java Code Executes?

The **execution** of a **Java application code** involves:





1- Object-Oriented Programming

Java is an object-oriented language, promoting the use of **objects** and **classes**. Organizing the program in the terms of a collection of objects is a way of object-oriented programming, each of which represents an instance of the class.

The **four main concepts of Object-Oriented programming** are:

- **Abstraction**
- **Encapsulation**
- **Inheritance**
- **Polymorphism**

2-Simplicity

Java's syntax is simple and easy to learn, especially for those familiar with C or C++. It eliminates complex features like pointers and multiple inheritances, making it easier to **write**, **debug**, and **maintain code**.

3-Security

In java, we don't have pointers, so we cannot access out-of-bound arrays i.e it shows ArrayIndexOutOfBoundsException if we try to do so. That's why several security flaws like stack corruption or buffer overflow are impossible to exploit in Java. Also, java programs run in an environment that is independent of the os(operating system) environment which makes java programs more secure.

4- Platform Independent

Compiler converts source code to byte code, and then the JVM executes the bytecode generated by the compiler. This byte code can run on any platform be it Windows, Linux, or macOS which means if we compile a program on Windows, then we can run it on Linux and vice versa. Each operating system has a different JVM, but the output produced by all the OS is the same after the execution of the byte code. That is why we call java a platform-independent language.

5- Robustness

Java language is robust which means reliable. It is developed in such a way that it puts a lot of effort into checking errors as early as possible, that is why the java compiler is able to detect even those errors that are not easy to detect by another programming language. The main features of java that make it robust are garbage collection, exception handling, and memory allocation.

6-Portability

As we know, java code written on one machine can be run on another machine. The platform-independent feature of java in which its platform-independent bytecode can be taken to any platform for execution makes java portable. WORA(Write Once Run Anywhere) makes java application to generate a '.class' file that corresponds to our applications(program) but contains code in binary format. It provides ease of architecture-neutral ease as bytecode is not dependent on any machine architecture. It is the primary reason java is used in the enterprising IT industry globally worldwide.

7-Architecture Neutral

The solution that the Java system adopts to solve the binary-distribution problem is a "binary code format" that's independent of hardware architectures, operating system interfaces, and window systems. The format of this system-independent binary code is architecture neutral.

8-Dynamic

Java is considered a dynamic language because of its ability to adapt and modify itself during runtime. This feature allows for greater flexibility in the coding process by enabling developers to update and alter elements of a program without requiring a complete recompilation.

9-Interpreted

Java can be considered both a compiled and an interpreted language because its source code is first compiled into a binary byte-code.

10-High Performance

Java architecture is defined in such a way that it reduces overhead during the runtime and at some times java uses Just In Time (JIT) compiler where the compiler compiles code on-demand basis where it only compiles those methods that are called making applications to execute faster.

11-Multithreading

Java supports multithreading, enabling the concurrent execution of multiple parts of a program. This feature is particularly useful for applications that require high performance, such as games and real-time simulations.

12-Distributed

We can create distributed applications using the java programming language. Remote Method Invocation and Enterprise Java Beans are used for creating distributed applications in java. The java programs can be easily distributed on one or more systems that are connected to each other through an internet connection.

2.2 My SQL

MySQL is an open-source relational database management system (RDBMS) that relies on structured query language (SQL) for accessing, managing, and interacting with databases. It's particularly known for its reliability, ease of use, and strong performance in handling large datasets, making it one of the most widely used database systems worldwide.

Here are some key points about MySQL:

- **Open Source:** Being open-source means that MySQL's source code is freely available, and it can be modified and distributed by anyone under the terms of its license.
- **Relational:** MySQL uses a relational model to store data in tables with rows and columns, which makes it easy to relate data across different tables using keys.
- **Cross-Platform:** MySQL is compatible with various operating systems, including Windows, Linux, and macOS.
- **Scalability:** It can handle large databases and supports numerous users simultaneously.
- **Security:** MySQL provides strong security features to protect sensitive data, including user account management, secure socket layer (SSL) connections, and encryption.

MySQL is commonly used in web applications and is a core component of the LAMP stack (Linux, Apache, MySQL, PHP/Perl/Python), which is a popular open-source web development platform.

Data Modelling –

Data Structure requirement according to the modules

- Identify the various Tables required.
- Fields for these tables.
- The various key Fields.
- Identify the various constraints.

-List of TABLES

1. Signup table
2. Signup2 table
3. Signup3 table
4. Login table
5. Bank table

1.Signup table:

FIELD	TYPE	CONSTRAINTS
form_no	Varchar(30)	Primary Key
Name	Varchar(30)	Not Null
father_name	Varchar(30)	Not Null

**

DOB	Varchar(30)	Not Null
gender	Varchar(30)	Not Null
email	Varchar(60)	Not Null
marital_status	Varchar(30)	Not Null
address	Varchar(60)	Not Null
city	Varchar(30)	Not Null
pincode	Varchar(30)	Not Null
state	Varchar(50)	Not Null

2.Signup2 table:

FIELD	TYPE	CONSTRAINTS
form_no	Varchar(30)	Primary Key
religion	Varchar(30)	Not Null
category	Varchar(30)	Not Null
income	Varchar(30)	Not Null
education	Varchar(30)	Not Null
occupation	Varchar(60)	Not Null
pan	Varchar(30)	Not Null
aadhar	Varchar(60)	Not Null
Senior citizen	Varchar(30)	Not Null
existing_account	Varchar(30)	Not Null

3.Signup3 table:

FIELD	TYPE	CONSTRAINTS
-------	------	-------------

**

form_no.	<u>Varchar(30)</u>	<u>Primary Key</u>
account_type	<u>Varchar(40)</u>	<u>Not Null</u>
card_number	<u>Varchar(30)</u>	<u>Not Null</u>
pin	<u>Varchar(30)</u>	<u>Not Null</u>
facility	<u>Varchar(200)</u>	<u>Not Null</u>

4.Login table:

FIELD	TYPE	CONSTRAINTS
form_no.	Varchar(30)	Primary Key
card_number	Varchar(50)	Not Null
pin	Varchar(30)	Not Null

5.Bank table:

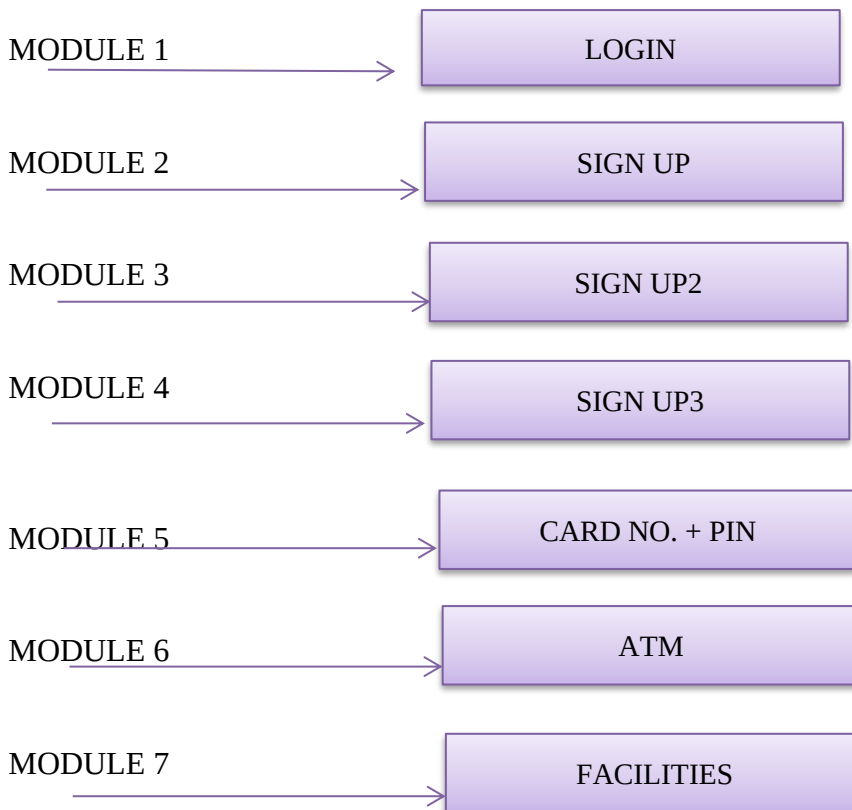
FIELD	TYPE	CONSTRAINTS
pin	Varchar(10)	Primary Key
date	Varchar(50)	Not Null
type	Varchar(20)	Not Null
amount	Varchar(20)	Not Null

CHAPTER 3

3.Designing

3.1Modules of the Project

- Module 1: Login = contain insertion for card number and pin with three options of sign in, sing up and clear.
- Module 2: Signup = contain page 1 of personal information form.
- Module 3: Signup2 = contain page 2 of additional information form.
- Module 4: Signup3 = contain page 3 of account details form.
- Module 5: generation of card number and pin after successful submit of forms.
- Module 6: access to ATM facility.
- Module 7: Transaction facilities: deposit, withdrawal of money, balance enquiry, fast cash, pin change and mini statement



3.2 ER Diagram

The Entity Relationship Model is a model for identifying entities to be represented in the database and representation of how those entities are related. The ER data model specifies enterprise schema that represents the overall logical structure of a database graphically.

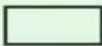
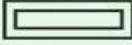
Peter Chen developed the ER diagram in 1976. The ER model was created to provide a simple and understandable model for representing the structure and logic of databases. It has since evolved into variations such as the Enhanced ER Model and the Object Relationship Model.

The Entity Relationship Diagram explains the relationship among the entities present in the database. ER models are used to model real-world objects like a person, a car, or a company and the relation between these real-world objects. In short, the ER Diagram is the structural format of the database. In this article we will see entity relationship model in detail.

Symbols Used in ER Model

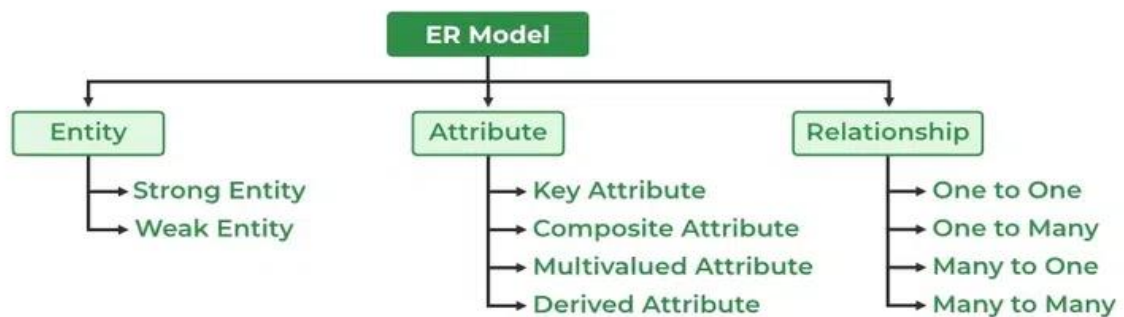
ER Model is used to model the logical view of the system from a data perspective which consists of these symbols:

- **Rectangles:** Rectangles represent Entities in the ER Model.
- **Ellipses:** Ellipses represent Attributes in the ER Model.
- **Diamond:** Diamonds represent Relationships among Entities.
- **Lines:** Lines represent attributes to entities and entity sets with other relationship types.
- **Double Ellipse:** Double Ellipses represent Multi-Valued Attributes.
- **Double Rectangle:** Double Rectangle represents a Weak Entity.

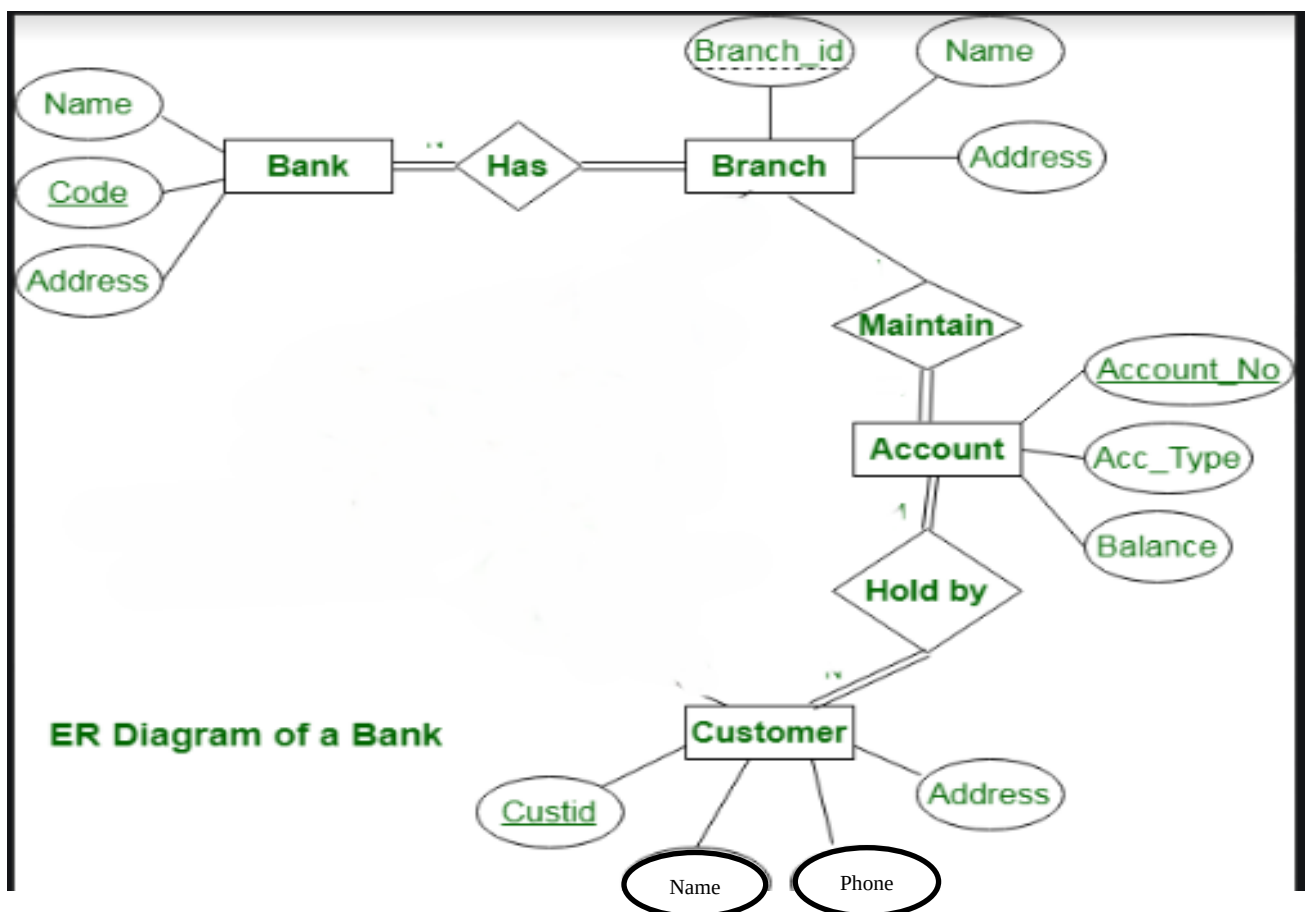
Figures	Symbols	Represents
Rectangle		Entities in ER Model
Ellipse		Attributes in ER Model
Diamond		Relationships among Entities
Line		Attributes to Entities and Entity Sets with Other Relationship Types
Double Ellipse		Multi-Valued Attributes
Double Rectangle		Weak Entity

Components of ER Diagram

ER Model consists of Entities, Attributes, and Relationships among Entities in a Database System.



E R Diagram of Bank



This bank ER diagram illustrates key information about bank, including entities such as branches, customers, accounts. It allows us to understand the relationships between entities.

- **Bank Entity** : Attributes of Bank Entity are Bank Name, Code and Address. Code is Primary Key for Bank Entity.
- **Customer Entity** : Attributes of Customer Entity are Customer_id, Name, Phone Number and Address. Customer_id is Primary Key for Customer Entity.
- **Branch Entity** : Attributes of Branch Entity are Branch_id, Name and Address. Branch_id is Primary Key for Branch Entity.
- **Account Entity** : Attributes of Account Entity are Account_number, Account_Type and Balance. Account_number is Primary Key for Account Entity.

3.3 DATA FLOW DIAGRAM

Data Flow Diagram (DFD) represents the flow of data within information systems. Data Flow Diagrams (DFD) provide a graphical representation of the data flow of a system that can be understood by both technical and non-technical users. The models enable software engineers, customers, and users to work together effectively during the analysis and specification of requirements.

Characteristics of Data Flow Diagram (DFD)

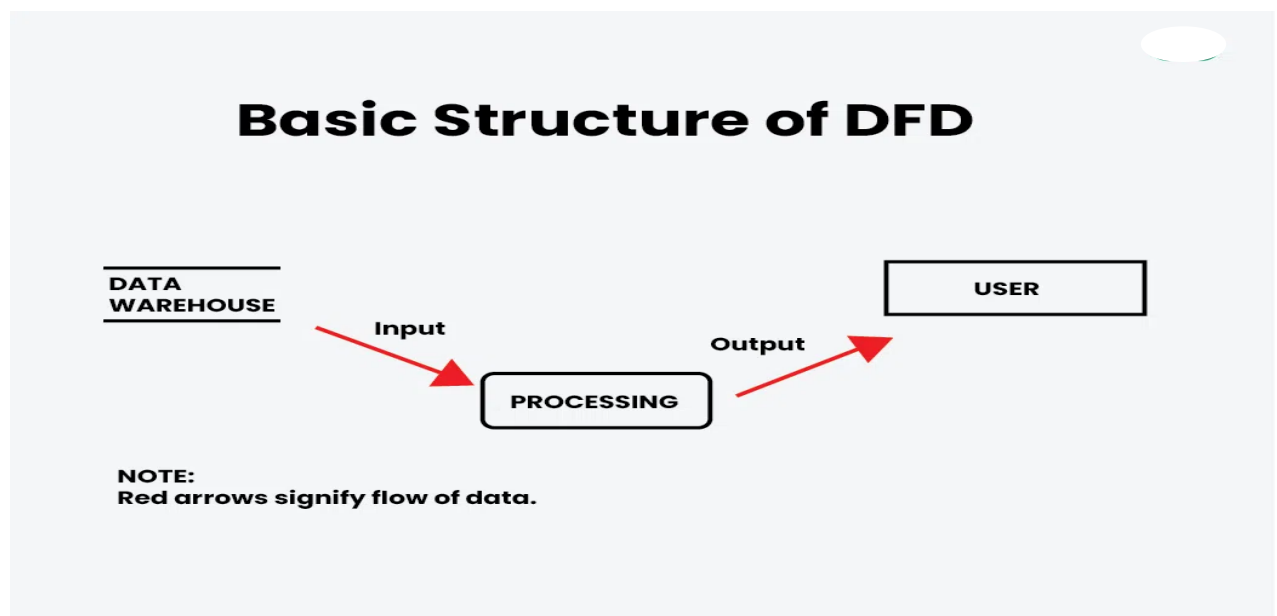
Below are some characteristics of Data Flow Diagram (DFD):

- **Graphical Representation:** Data Flow Diagram (DFD) use different symbols and notation to represent data flow within system. That simplify the complex model.
- **Problem Analysis:** Data Flow Diagram (DFDs) are very useful in understanding a system and can be effectively used during analysis. Data Flow Diagram (DFDs) are quite general and are not limited to problem analysis for software requirements specification.
- **Abstraction:** Data Flow Diagram (DFD) provides a abstraction to complex model i.e. DFD hides unnecessary implementation details and show only the flow of data and processes within information system.
- **Hierarchy:** Data Flow Diagram (DFD) provides a hierarchy of a system. High- level diagram i.e. 0-level diagram provides an overview of entire system while lower-level diagram like 1-level DFD and beyond provides a detailed data flow of individual process.
- **Data Flow:** The primary objective of Data Flow Diagram (DFD) is to visualize the data flow between external entity, processes and data store. Data Flow is represented by an arrow Symbol.
- **Ease of Understanding:** Data Flow Diagram (DFD) can be easily understand by both technical and non-technical stakeholders.
- **Modularity:** Modularity can be achieved using Data Flow Diagram (DFD) as it breaks the complex system into smaller module or processes. This provides easily analysis and design of a system.

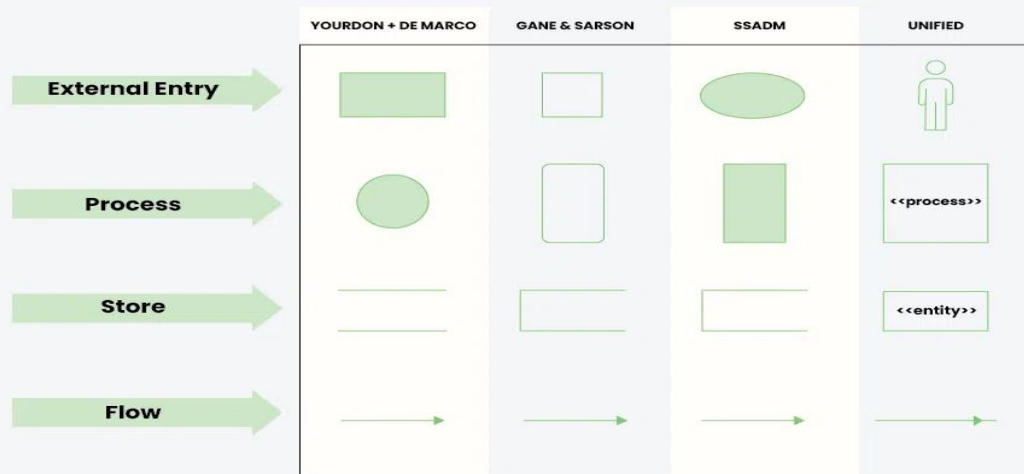
Components of Data Flow Diagrams (DFD)

The Data Flow Diagram has 4 components:

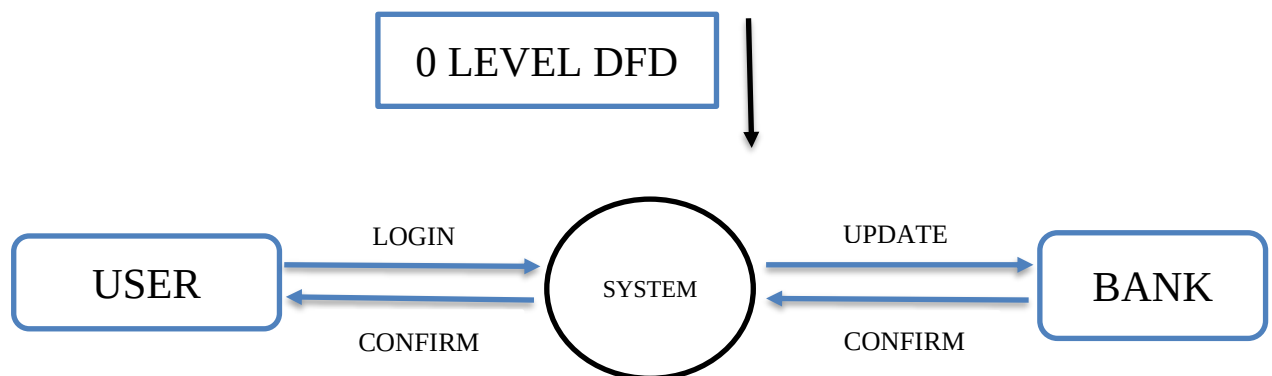
- **Process:** Input to output transformation in a system takes place because of process function. The symbols of a process are rectangular with rounded corners, oval, rectangle or a circle. The process is named a short sentence, in one word or a phrase to express its essence
- **Data Flow:** Data flow describes the information transferring between different parts of the systems. The arrow symbol is the symbol of data flow. A relatable name should be given to the flow to determine the information which is being moved. Data flow also represents material along with information that is being moved. Material shifts are modeled in systems that are not merely informative. A given flow should only transfer a single type of information. The direction of flow is represented by the arrow which can also be bi-directional.
- **Warehouse (Data Store) :** The data is stored in the warehouse for later use. Two horizontal lines represent the symbol of the store. The warehouse is simply not restricted to being a data file rather it can be anything like a folder with documents, an optical disc, a filing cabinet. The data warehouse can be viewed independent of its implementation. When the data flow from the warehouse it is considered as data reading and when data flows to the warehouse it is called data entry or data updating.
- **Terminator (External Entity):** The Terminator is an external entity that stands outside of the system and communicates with the system. It can be, for example, organizations like banks, groups of people like customers or different departments of the same organization, which is not a part of the model system and is an external entity. Modeled systems also communicate with terminator.



Data Flow Diagram Methods & Symbols



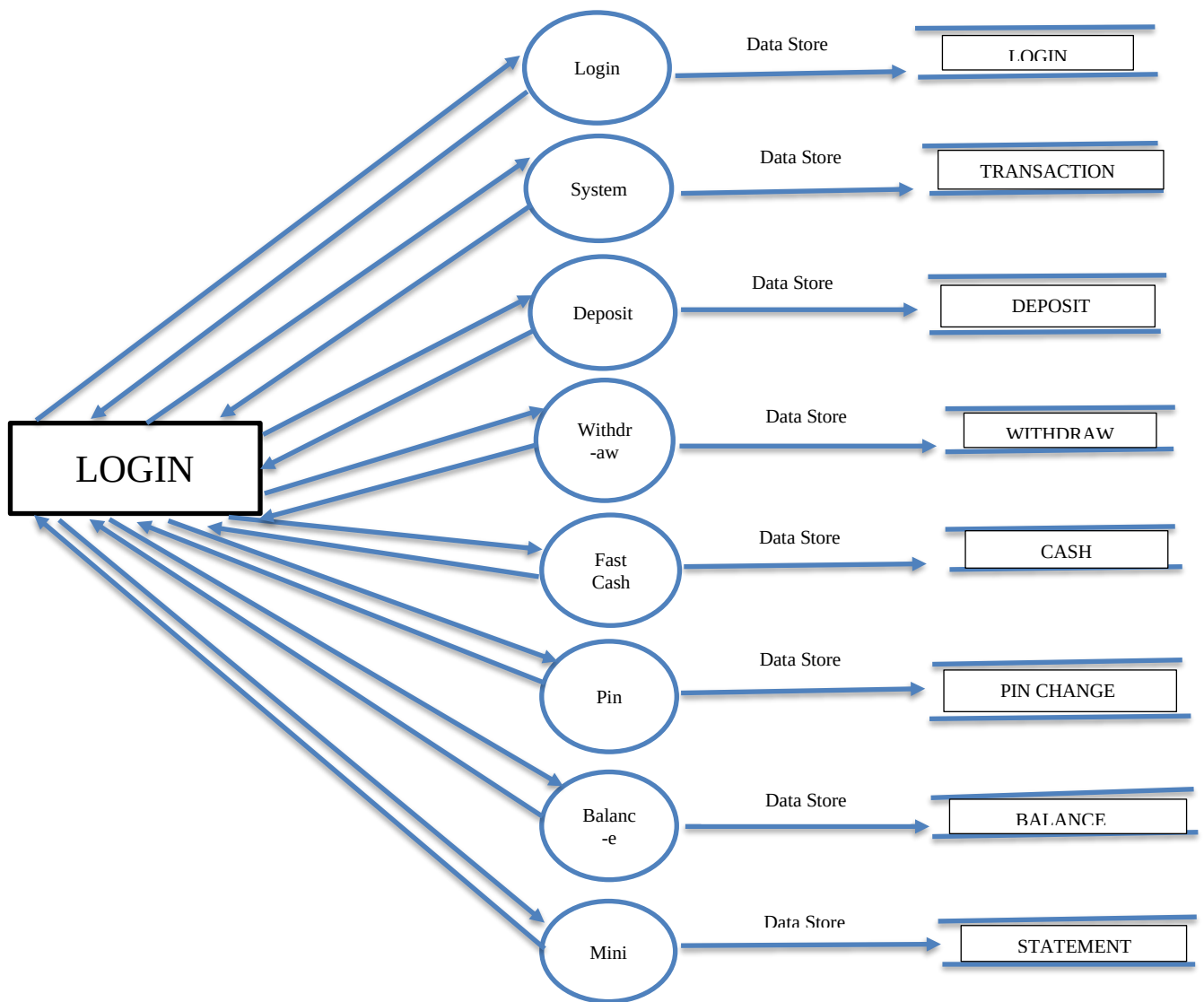
0-LEVEL DFD(Context diagram)



It is also known as a context diagram. It's designed to be an abstraction view, showing the system as a single process with its relationship to external entities. It represents the entire system as a single bubble with input and output data indicated by incoming/outgoing arrows.

1-LEVEL DFD

This level provides a more detailed view of the system by breaking down the major processes identified in the level 0 DFD into sub-processes. Each sub-process is depicted as a separate process on the level 1 DFD. The data flows and data stores associated with each sub-process are also shown. In 1-level DFD, the context diagram is decomposed into multiple bubbles/processes. In this level, we highlight the main functions of the system and breakdown the high-level process of 0-level DFD into subprocesses.

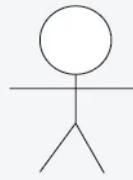


3.4 Use Case Diagram

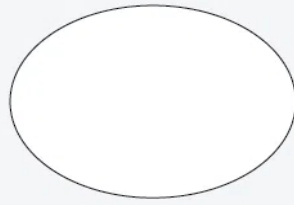
A Use Case Diagram in Unified Modeling Language (UML) is a visual representation that illustrates the interactions between users (actors) and a system. It captures the functional requirements of a system, showing how different users engage with various use cases, or specific functionalities, within the system. Use case diagrams provide a high-level overview of a system's behavior, making them useful for stakeholders, developers, and analysts to understand how a system is intended to operate from the user's perspective, and how different processes relate to one another. They are crucial for defining system scope and requirements.

A Use Case Diagram is a type of Unified Modeling Language (UML) diagram that represents the interaction between actors (users or external systems) and a system under consideration to accomplish specific goals. It provides a high-level view of the system's functionality by illustrating the various ways users can interact with it.

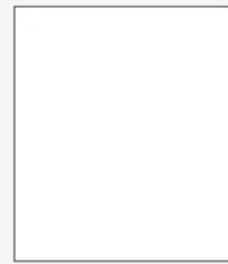
Use Case Diagram Notations



Actor



UseCase



System Boundary

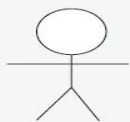
Use Case Diagrams | Unified Modeling Language (UML)



1. Actors

Actors are external entities that interact with the system. These can include users, other systems, or hardware devices. In the context of a Use Case Diagram, actors initiate use cases and receive the outcomes. Proper identification and understanding of actors are crucial for accurately modeling system behavior.

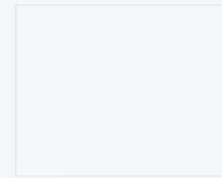
Actor



Actors are typically represented by stick figures



Use cases are represented by ovals.



System boundary is represented by a rectangular box.

Use Case Diagrams | Unified Modeling Language (UML)



2. Use Cases

Use cases are like scenes in the play. They represent specific things your system can do. In the online shopping system, examples of use cases could be "Place Order," "Track Delivery," or "Update Product Information". Use cases are represented by ovals.

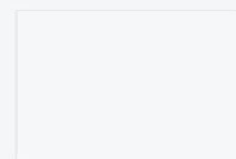
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Actors are typically represented by stick figures



Use cases are represented by ovals.



System boundary is represented by a rectangular box.

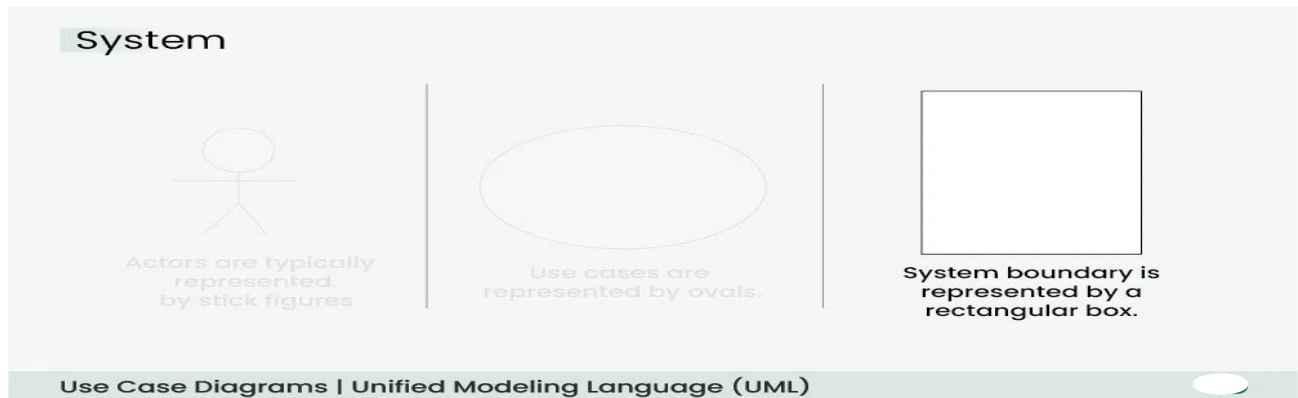
Use Case Diagrams | Unified Modeling Language (UML)



**

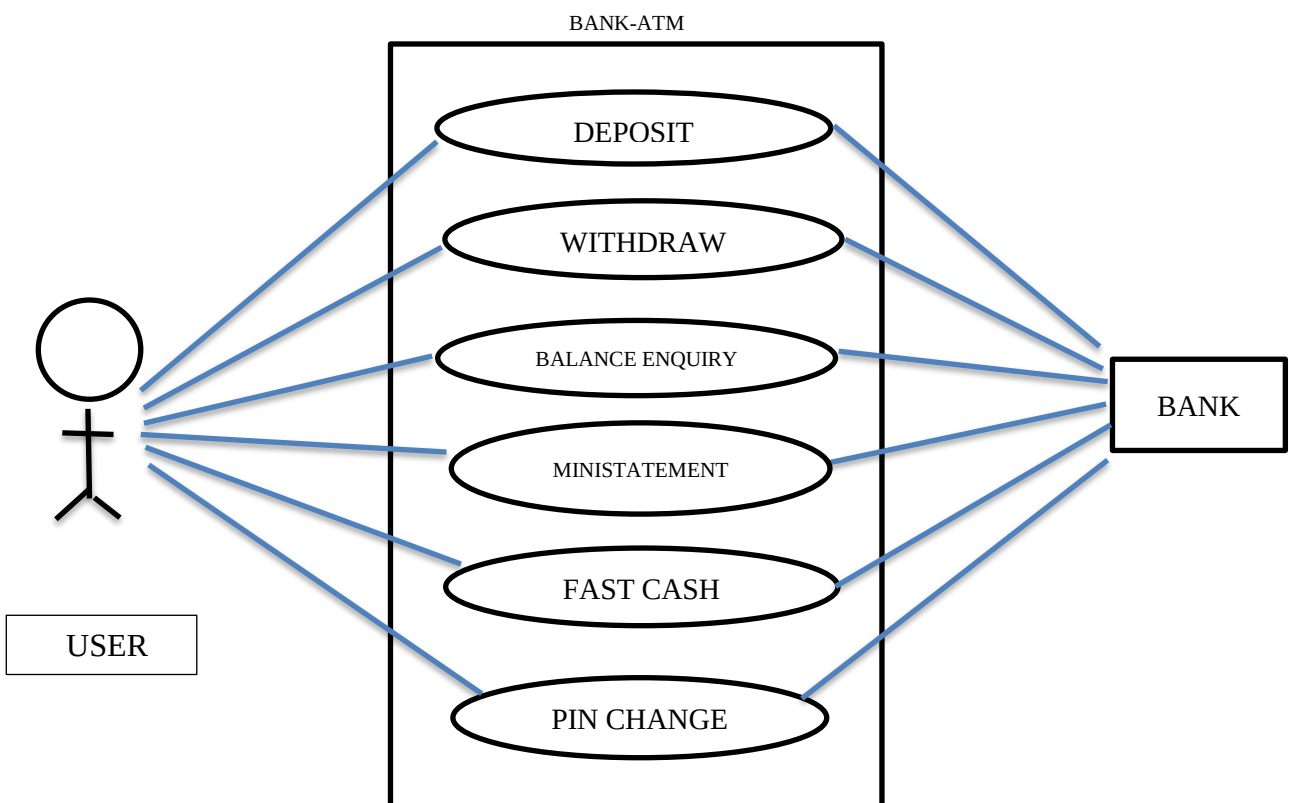
3. System Boundary

The system boundary is a visual representation of the scope or limits of the system you are modeling. It defines what is inside the system and what is outside. The boundary helps to establish a clear distinction between the elements that are part of the system and those that are external to it. The system boundary is typically represented by a rectangular box that surrounds all the use cases of the system.



The purpose of system boundary is to clearly outlines the boundaries of the system, indicating which components are internal to the system and which are external actors or entities interacting with the system.

Use Case Diagram



CHAPTER 4

4.1 MINIMUM HARDWARE REQUIRED

- PROCESSOR = DUAL CORE
- HARD DISC = 50 GB
- MEMORY = 1 GB RAM

IntelliJ IDE

IntelliJ IDEA is a powerful Integrated Development Environment (IDE) developed by JetBrains, primarily designed for Java and Kotlin development. It offers a wide range of features that enhance productivity and streamline the development process. Here are some key aspects of IntelliJ IDEA:

1. Smart Code Completion: IntelliJ IDEA provides intelligent code completion, which helps developers write code faster by suggesting relevant methods, variables, and keywords as they type.
2. Code Navigation: The IDE offers advanced code navigation features, allowing developers to quickly jump to any class, method, or file within the project.
3. Refactoring Tools: IntelliJ IDEA includes powerful refactoring tools that help developers improve and optimize their code without changing its external behavior.
4. Version Control Integration: The IDE seamlessly integrates with popular version control systems like Git, SVN, and Mercurial, making it easy to manage code changes and collaborate with team members.
5. Debugging and Testing: IntelliJ IDEA provides robust debugging and testing tools, including a debugger, test runners, and code coverage analysis.
6. Plugins and Extensibility: The IDE supports a wide range of plugins and extensions, allowing developers to customize their development environment to suit their specific needs.
7. Cross-Platform Support: IntelliJ IDEA is available on Windows, macOS, and Linux, making it accessible to developers on different operating systems.



4.2.1 JAVA JDK 23 = FRONT END

Java Development Kit (JDK) 23 is a long-awaited release, which brings numerous **new features, enhancements** and **updates** to increase **performance, security** and the overall experience of **developers**. This guide wants to provide an extensive review of what will be in **JDK 23** such as **new characteristics**, necessary additions, upgrades from **JDK 22** and the current state of releasing. Since **JDK 23** is currently in **Initial Release Candidate**, these features are expected but not yet finalized. This release has been built for **Java programming** with advanced **performance optimizations**, cutting-edge **security measures** and improved tool support.

By focusing on **modern programming paradigms**, **JDK 23** ensures that **Java** remains competitive and relevant in today's rapidly evolving technology landscape. Whether you've been developing in **Java** for years or you're just getting started, it's important to understand the **updates and improvements** that come with **JDK 23** so as to harness its full potential and stay ahead in the **software development** industry. With this upgrade, **developers** have made significant strides in productivity, surpassing those who are slower when writing code. This results in better, more efficient programs written in **Java**, making **JDK 23** a must-have enhancement for every project based on **Java**.

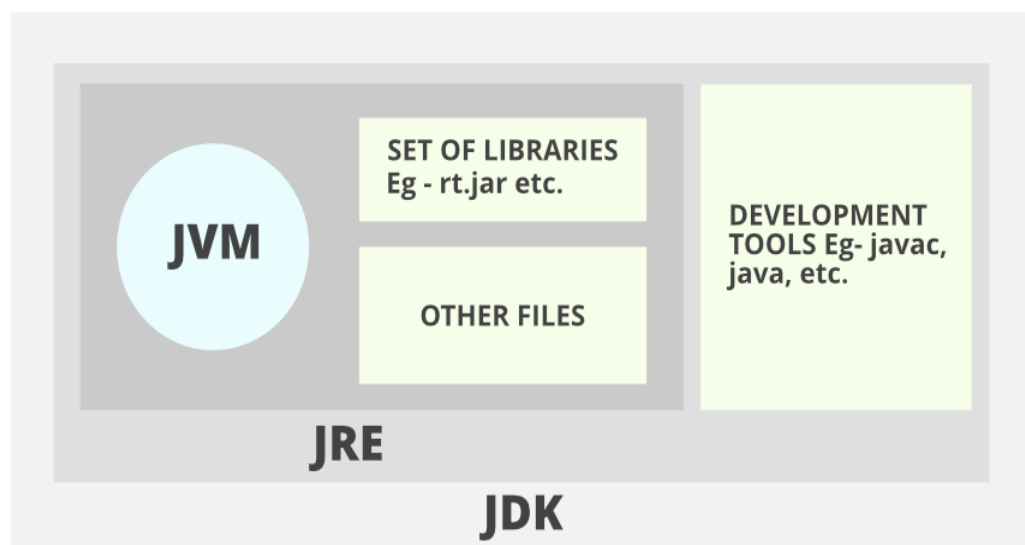
1. **JDK** (Java Development Kit) is a Kit that provides the environment to develop and execute(run) the Java program. **JDK** is a kit(or package) that includes two things

Development Tools(to provide an environment to develop your java programs)

JRE (to execute your java program).

2. **JRE** (Java Runtime Environment) is an installation package that provides an environment to only run(not develop) the java program(or application)onto your machine. **JRE** is only used by those who only want to run Java programs that are end-users of your system.

3. **JVM** (Java Virtual Machine) is a very important part of both **JDK** and **JRE** because it is contained or inbuilt in both. Whatever Java program you run using **JRE** or **JDK** goes into **JVM** and **JVM** is responsible for executing the java program line by line, hence it is also known as an **interpreter**.



4.2.2 My SQL = Backend

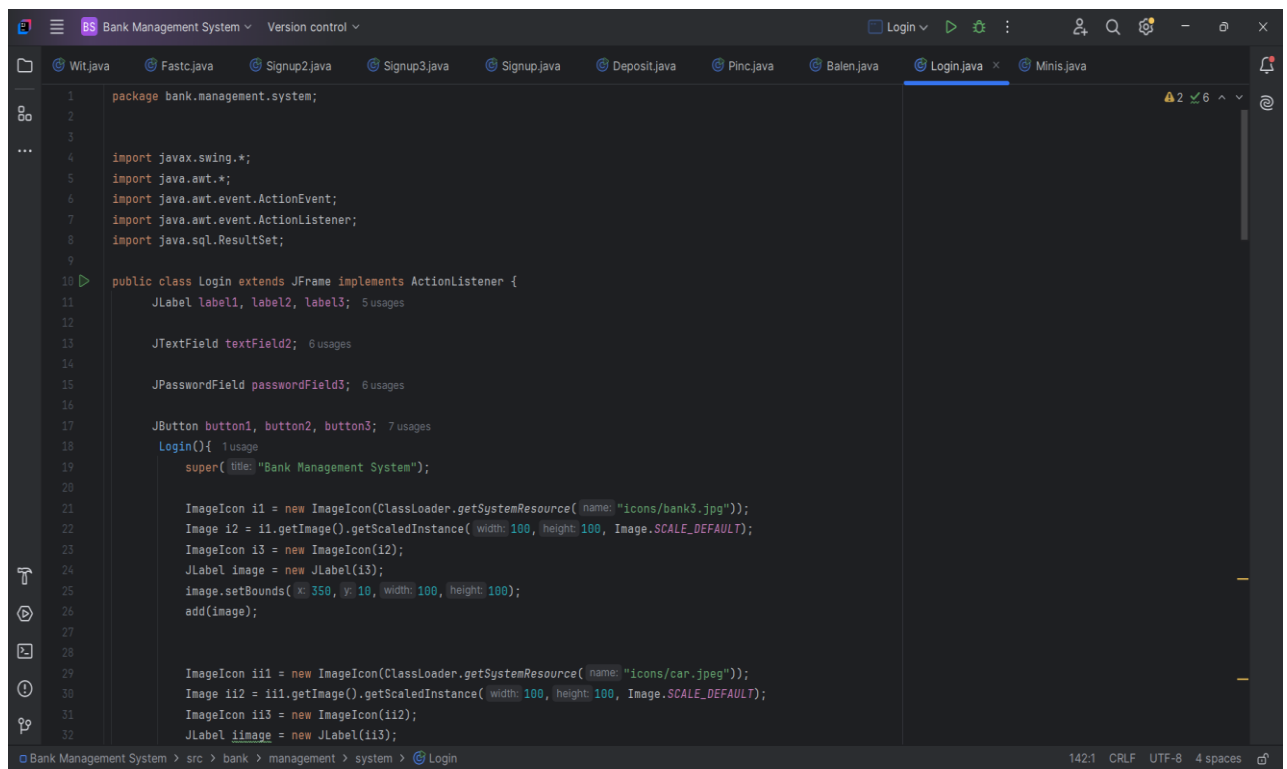
MySQL is a popular choice for managing **relational databases** for several reasons:

1. **Open Source:** MySQL is open-source software, which means it's **free to use** and has a large community of developers contributing to its improvement.
2. **Relational:** MySQL follows the relational database model, allowing users to organize data into **tables** with **rows** and **columns**, facilitating efficient **data storage** and retrieval.
3. **Reliability:** MySQL has been around for a long time and is known for its **stability** and **reliability**.
4. **Performance:** MySQL is optimized for performance, making it capable of handling **high-volume transactions** and large datasets efficiently.
5. **Scalability:** MySQL can scale both **vertically** and **horizontally** to accommodate growing data and user loads. You can add more resources to a single server or distribute the workload across multiple servers using techniques like sharding or replication.
6. **Compatibility:** MySQL is widely supported by many **programming languages**, frameworks, and tools. It offers connectors and APIs for popular languages like PHP, Python, Java, and more, making it easy to integrate with your existing software stack.
7. **Security:** MySQL provides robust **security features** to protect your data, including access controls, encryption, and auditing capabilities. With proper configuration, you can ensure that only authorized users have access to sensitive information.

0

4.3 JAVA Programming

Login.java



```
1 package bank.management.system;
2
3
4 import javax.swing.*;
5 import java.awt.*;
6 import java.awt.event.ActionEvent;
7 import java.awt.event.ActionListener;
8 import java.sql.ResultSet;
9
10 public class Login extends JFrame implements ActionListener {
11     JLabel label1, label2, label3; 5 usages
12
13     JTextField textField2; 6 usages
14
15     JPasswordField passwordField3; 6 usages
16
17     JButton button1, button2, button3; 7 usages
18     Login(){ 1 usage
19         super("Bank Management System");
20
21         ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource("icons/bank3.jpg"));
22         Image i2 = i1.getImage().getScaledInstance(100, 100, Image.SCALE_DEFAULT);
23         ImageIcon i3 = new ImageIcon(i2);
24         JLabel image = new JLabel(i3);
25         image.setBounds(350, 10, 100, 100);
26         add(image);
27
28
29         ImageIcon i11 = new ImageIcon(ClassLoader.getSystemResource("icons/car.jpeg"));
30         Image i12 = i11.getImage().getScaledInstance(100, 100, Image.SCALE_DEFAULT);
31         ImageIcon i13 = new ImageIcon(i12);
32         JLabel iimage = new JLabel(i13);
```

**

```
Bank Management System  Version control  Login  Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java

18 public class Login extends JFrame implements ActionListener {
18     Login(){ 1 usage
33         iimage.setBounds( x: 630, y: 350, width: 100, height: 100);
34         add(iimage);
35
36         label1 = new JLabel( text: "WELCOME TO ATM");
37         label1.setForeground(Color.BLACK);
38         label1.setFont(new Font( name: "AvantGarde", Font.BOLD, size: 38));
39         label1.setBounds( x: 230, y: 125, width: 450, height: 40);
40         add(label1);
41
42
43
44         label2 = new JLabel( text: "Card No. :");
45         label2.setForeground(Color.BLACK);
46         label2.setFont(new Font( name: "Releway", Font.BOLD, size: 28));
47         label2.setBounds( x: 150, y: 190, width: 375, height: 30);
48         add(label2);
49
50         textfield2 = new JTextField( columns: 15);
51         textfield2.setBounds( x: 325, y: 190, width: 230, height: 30);
52         textfield2.setFont(new Font( name: "Arial", Font.BOLD, size: 14));
53         add(textfield2);
54
55
56         label3 = new JLabel( text: "PIN :");
57         label3.setForeground(Color.BLACK);
58         label3.setFont(new Font( name: "Releway", Font.BOLD, size: 28));
59         label3.setBounds( x: 150, y: 250, width: 375, height: 30);
60         add(label3);
61
62         passwordField3 = new JPasswordField( columns: 15);
```

Bank Management System > src > bank > management > system > Login 142:1 CRLF UTF-8 4 spaces

```
Bank Management System  Version control  Login  Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java

18 public class Login extends JFrame implements ActionListener {
18     Login(){ 1 usage
60         add(label3);
61
62         passwordField3 = new JPasswordField( columns: 15);
63         passwordField3.setBounds( x: 325, y: 250, width: 230, height: 30);
64         passwordField3.setFont(new Font( name: "Arial", Font.BOLD, size: 14));
65         add(passwordField3);
66
67         button1 = new JButton( text: "SIGN IN");
68         button1.setFont(new Font( name: "Arial", Font.BOLD, size: 14));
69         button1.setForeground(Color.BLACK);
70         button1.setBounds( x: 325, y: 300, width: 100, height: 30);
71         button1.addActionListener( this);
72         add(button1);
73
74
75         button2 = new JButton( text: "CLEAR");
76         button2.setFont(new Font( name: "Arial", Font.BOLD, size: 14));
77         button2.setForeground(Color.BLACK);
78         button2.setBounds( x: 455, y: 300, width: 100, height: 30);
79         button2.addActionListener( this);
80         add(button2);
81
82
83         button3 = new JButton( text: "SIGN UP");
84         button3.setFont(new Font( name: "Arial", Font.BOLD, size: 14));
85         button3.setForeground(Color.BLACK);
86         button3.setBounds( x: 325, y: 350, width: 230, height: 30);
87         button3.addActionListener( this);
88         add(button3);
89
```

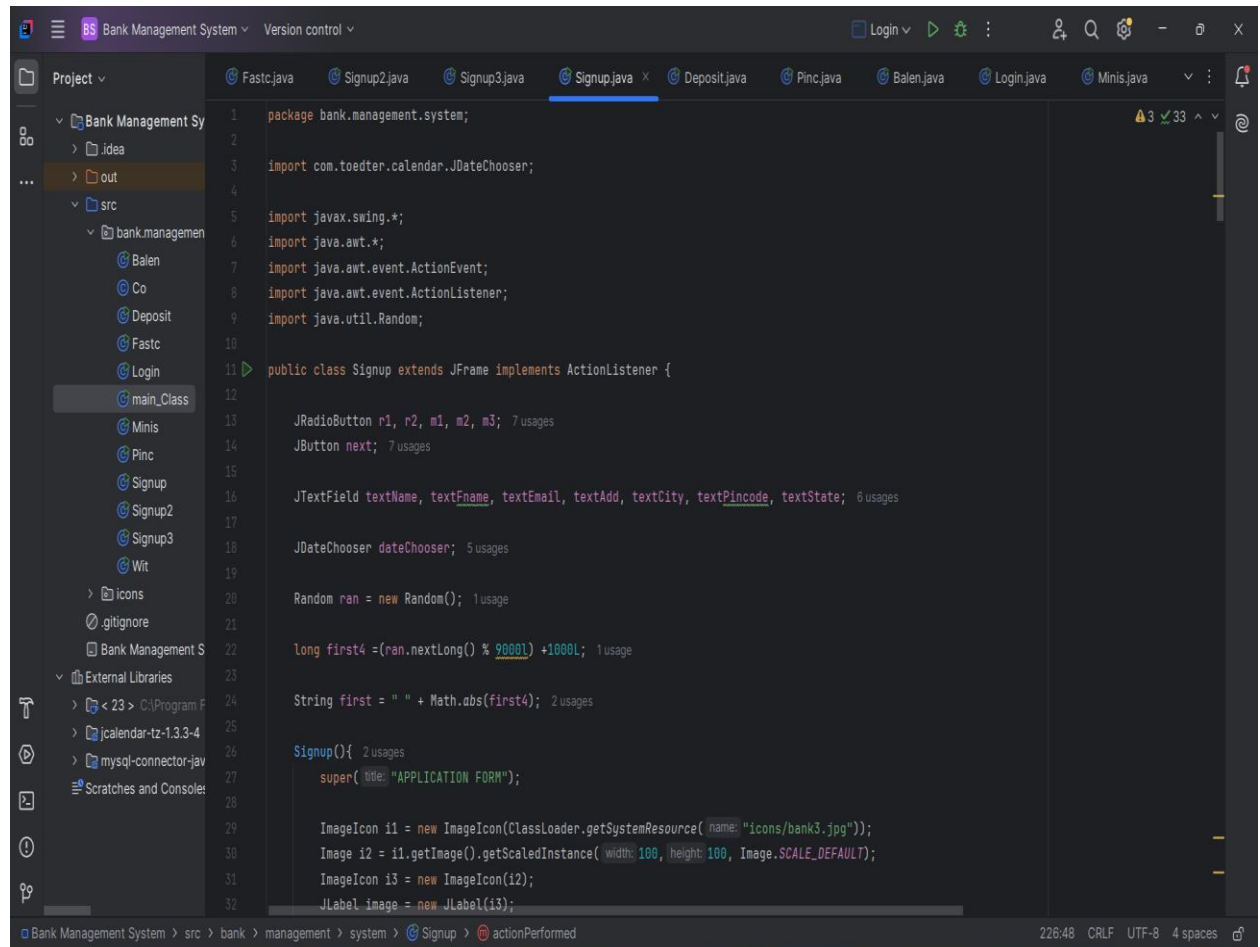
Bank Management System > src > bank > management > system > Login 142:1 CRLF UTF-8 4 spaces

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Bank Management System Version control
Login
Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java
10 public class Login extends JFrame implements ActionListener {
11     Login() {
12         ImageIcon iiii1 = new ImageIcon(ClassLoader.getResource("icons/backup.jpg"));
13         Image iiii2 = iiii1.getImage().getScaledInstance(850, 480, Image.SCALE_DEFAULT);
14         ImageIcon iiii3 = new ImageIcon(iiii2);
15         JLabel iiii4 = new JLabel(iiii3);
16         iiii4.setBounds(0, 0, 850, 480);
17         add(iiii4);
18
19         setLayout(null);
20         setSize(850, 480);
21         setLocation(450, 200);
22         setVisible(true);
23     }
24
25     @Override
26     public void actionPerformed(ActionEvent e) {
27         try {
28             if (e.getSource() == button1) {
29                 Co c = new Co();
30                 String cardno = textField2.getText();
31                 String pin = passwordField3.getText();
32                 String q = "select * from login where card_number = '" + cardno + "' and 'pin = '" + pin + "'";
33                 ResultSet resultSet = c.statement.executeQuery(q);
34                 if (resultSet.next()) {
35                     setVisible(false);
36                     new main_Class(pin);
37                 } else {
38                     JOptionPane.showMessageDialog(parentComponent, null, message: "Incorrect Card Number or Pin");
39                 }
40             }
41         } catch (Exception e1) {
42             e1.printStackTrace();
43         }
44     }
45 }
142:1 CRLF UTF-8 4 spaces
```

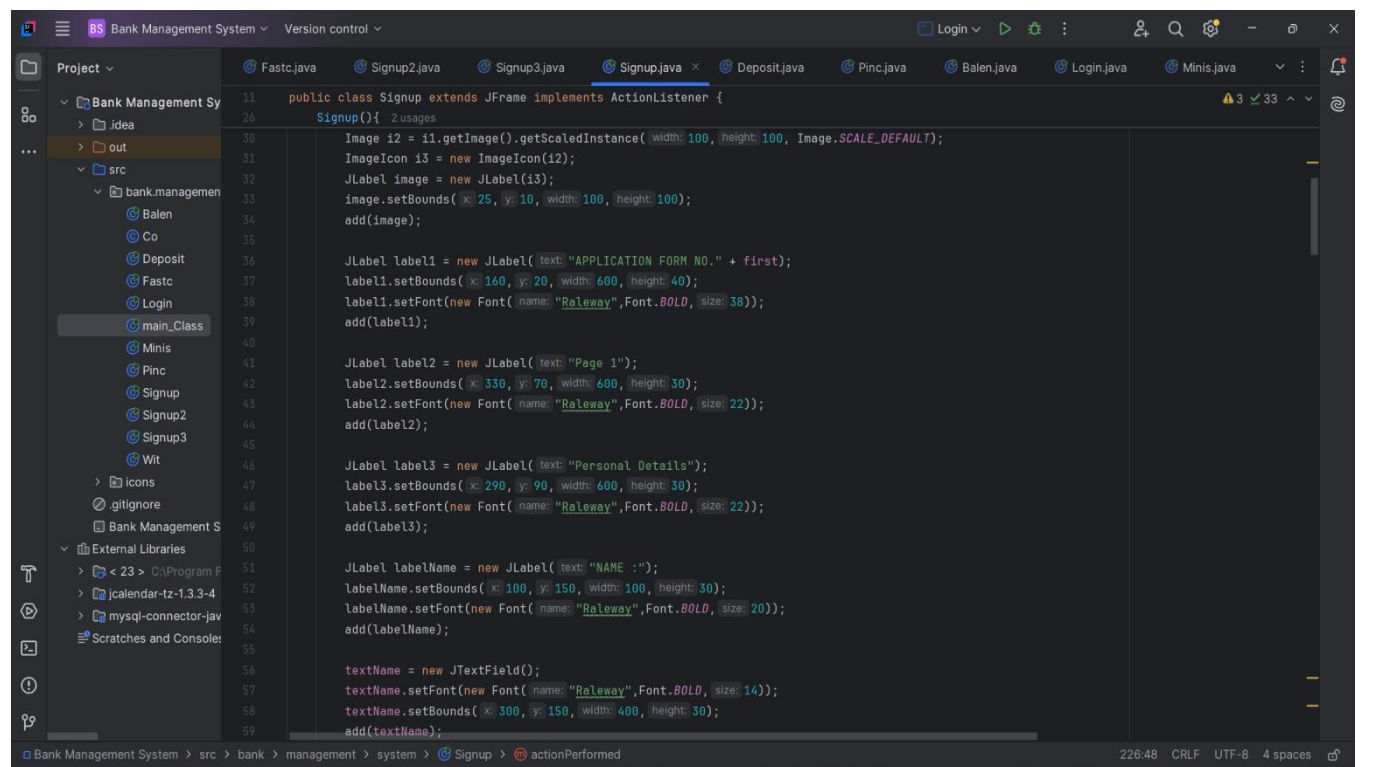
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11     public void actionPerformed(ActionEvent e) {
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```

Signup.java



```
1 package bank.management.system;
2
3 import com.toedter.calendar.JDateChooser;
4
5 import javax.swing.*;
6 import java.awt.*;
7 import java.awt.event.ActionEvent;
8 import java.awt.event.ActionListener;
9 import java.util.Random;
10
11 public class Signup extends JFrame implements ActionListener {
12
13     JRadioButton r1, r2, m1, m2, m3;
14     JButton next;
15
16     JTextField textName, textFname, textEmail, textAdd, textCity, textPincode, textState;
17
18     JDateChooser dateChooser;
19
20     Random ran = new Random();
21
22     long first4 = (ran.nextLong() % 9000L) + 1000L;
23
24     String first = " " + Math.abs(first4);
25
26     Signup() {
27         super("APPLICATION FORM");
28
29         ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource("icons/bank3.jpg"));
30         Image i2 = i1.getImage().getScaledInstance(100, 100, Image.SCALE_DEFAULT);
31         ImageIcon i3 = new ImageIcon(i2);
32         JLabel image = new JLabel(i3);
```

2



```
33         image.setBounds(25, 10, 100, 100);
34         add(image);
35
36         JLabel label1 = new JLabel("APPLICATION FORM NO. " + first);
37         label1.setBounds(160, 20, 600, 40);
38         label1.setFont(new Font("Raleway", Font.BOLD, 38));
39         add(label1);
40
41         JLabel label2 = new JLabel("Page 1");
42         label2.setBounds(330, 70, 600, 30);
43         label2.setFont(new Font("Raleway", Font.BOLD, 22));
44         add(label2);
45
46         JLabel label3 = new JLabel("Personal Details");
47         label3.setBounds(290, 90, 600, 30);
48         label3.setFont(new Font("Raleway", Font.BOLD, 22));
49         add(label3);
50
51         JLabel labelName = new JLabel("NAME :");
52         labelName.setBounds(100, 150, 100, 30);
53         labelName.setFont(new Font("Raleway", Font.BOLD, 20));
54         add(labelName);
55
56         textName = new JTextField();
57         textName.setFont(new Font("Raleway", Font.BOLD, 14));
58         textName.setBounds(300, 150, 400, 30);
59         add(textName);
```

**

```
11 public class Signup extends JFrame implements ActionListener {
26     Signup(){
57         textName.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
58         textName.setBounds( x: 300, y: 150, width: 400, height: 30);
59         add(textName);
60
61         JLabel labelFname = new JLabel( text: "FATHER'S NAME :");
62         labelFname.setBounds( x: 100, y: 200, width: 200, height: 30);
63         labelFname.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 20));
64         add(labelFname);
65
66         textFname = new JTextField();
67         textFname.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
68         textFname.setBounds( x: 300, y: 200, width: 400, height: 30);
69         add(textFname);
70
71         JLabel labelDOB = new JLabel( text: "DATE OF BIRTH :");
72         labelDOB.setBounds( x: 100, y: 250, width: 200, height: 30);
73         labelDOB.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 20));
74         add(labelDOB);
75
76         dateChooser = new JDateChooser();
77         dateChooser.setForeground(new Color( r: 150, g: 150, b: 150));
78         dateChooser.setBounds( x: 300, y: 250, width: 400, height: 30);
79         add(dateChooser);
80
81         JLabel labelG = new JLabel( text: "GENDER :");
82         labelG.setBounds( x: 100, y: 300, width: 200, height: 30);
83         labelG.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 20));
84         add(labelG);
85
86         r1 = new JRadioButton( text: "MALE");
```

```
87         r1.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
88         r1.setBackground(new Color( r: 148, g: 203, b: 242));
89         r1.setBounds( x: 300, y: 300, width: 70, height: 30);
90         add(r1);
91
92         r2 = new JRadioButton( text: "FEMALE");
93         r2.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
94         r2.setBackground(new Color( r: 148, g: 203, b: 242));
95         r2.setBounds( x: 450, y: 300, width: 90, height: 30);
96         add(r2);
97
98         ButtonGroup buttonGroup = new ButtonGroup();
99         buttonGroup.add(r1);
100        buttonGroup.add(r2);
101
102        JLabel labelEmail = new JLabel( text: "EMAIL ADDRESS :");
103        labelEmail.setBounds( x: 100, y: 350, width: 200, height: 30);
104        labelEmail.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 20));
105        add(labelEmail);
106
107        textEmail = new JTextField();
108        textEmail.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
109        textEmail.setBounds( x: 300, y: 350, width: 400, height: 30);
110        add(textEmail);
111
112        JLabel labelMs = new JLabel( text: "MARITAL STATUS :");
113        labelMs.setBounds( x: 100, y: 400, width: 200, height: 30);
114        labelMs.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 20));
115        add(labelMs);
116
```

**


```
Bank Management System  Version control  Login  Run  Debug  Search  Settings  Window  Help  X

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External Libraries
  < 23 > C:\Program F
  jcalendar-tz-1.3.3-4
  mysql-connector-jav
  Scratches and Console:

11  public class Signup extends JFrame implements ActionListener {
26      Signup(){ 2 usages
117          m1 = new JRadioButton( text: "MARRIED");
118          m1.setFont(new Font( name: "RaLeway",Font.BOLD, size: 14));
119          m1.setBackground(new Color( r: 148, g: 203, b: 242));
120          m1.setBounds( x: 300, y: 400, width: 100, height: 30);
121          add(m1);
122
123          m2 = new JRadioButton( text: "UNMARRIED");
124          m2.setFont(new Font( name: "RaLeway",Font.BOLD, size: 14));
125          m2.setBackground(new Color( r: 148, g: 203, b: 242));
126          m2.setBounds( x: 450, y: 400, width: 120, height: 30);
127          add(m2);
128
129          m3 = new JRadioButton( text: "OTHER");
130          m3.setFont(new Font( name: "RaLeway",Font.BOLD, size: 14));
131          m3.setBackground(new Color( r: 148, g: 203, b: 242));
132          m3.setBounds( x: 620, y: 400, width: 80, height: 30);
133          add(m3);
134
135          ButtonGroup buttonGroup1 = new ButtonGroup();
136          buttonGroup1.add(m1);
137          buttonGroup1.add(m2);
138          buttonGroup1.add(m3);
139
140          JLabel labelAdd = new JLabel( text: "ADDRESS :");
141          labelAdd.setBounds( x: 100, y: 450, width: 200, height: 30);
142          labelAdd.setFont(new Font( name: "RaLeway",Font.BOLD, size: 20));
143          add(labelAdd);
144
145          textAdd = new JTextField();
146
```

```
Bank Management System  Version control  Login  Run  Debug  Search  Settings  Window  Help  X

Project  Fastc.java  Signup2.java  Signup3.java  Signup.java x  Deposit.java  Pinc.java  Balen.java  Login.java  Minis.java  ...
Bank Management System
  .idea
  out
  src
    bank.managemen
      Balen
      Co
      Deposit
      Fastc
      Login
      main_Class
      Minis
      Pinc
      Signup
      Signup2
      Signup3
      Wit
    icons
    .gitignore
  Bank Management S
External Libraries
  < 23 > C:\Program F
  jcalendar-tz-1.3.3-4
  mysql-connector-jav
  Scratches and Console:

11  public class Signup extends JFrame implements ActionListener {
26      Signup(){ 2 usages
147          textAdd.setFont(new Font( name: "RaLeway",Font.BOLD, size: 14));
148          textAdd.setBounds( x: 300, y: 450, width: 400, height: 30);
149          add(textAdd);
150
151          JLabel labelCity = new JLabel( text: "CITY :");
152          labelCity.setBounds( x: 100, y: 500, width: 200, height: 30);
153          labelCity.setFont(new Font( name: "RaLeway",Font.BOLD, size: 20));
154          add(labelCity);
155
156          textCity = new JTextField();
157          textCity.setFont(new Font( name: "RaLeway",Font.BOLD, size: 14));
158          textCity.setBounds( x: 300, y: 500, width: 400, height: 30);
159          add(textCity);
160
161          JLabel labelPin = new JLabel( text: "PINCODE :");
162          labelPin.setBounds( x: 100, y: 550, width: 200, height: 30);
163          labelPin.setFont(new Font( name: "RaLeway",Font.BOLD, size: 20));
164          add(labelPin);
165
166          textPincode = new JTextField();
167          textPincode.setFont(new Font( name: "RaLeway",Font.BOLD, size: 14));
168          textPincode.setBounds( x: 300, y: 550, width: 400, height: 30);
169          add(textPincode);
170
171          JLabel labelState = new JLabel( text: "STATE :");
172          labelState.setBounds( x: 100, y: 600, width: 200, height: 30);
173          labelState.setFont(new Font( name: "RaLeway",Font.BOLD, size: 20));
174          add(labelState);
175
176
```

**

```
11 public class Signup extends JFrame implements ActionListener {
26     Signup(){
177         textState = new JTextField();
178         textState.setFont(new Font( name: "Raleway",Font.BOLD, size: 14));
179         textState.setBounds( x: 300, y: 600, width: 400, height: 30);
180         add(textState);
181
182         next = new JButton( text: "NEXT");
183         next.setFont(new Font( name: "Raleway", Font.BOLD, size: 18));
184         next.setBackground(Color.BLACK);
185         next.setForeground(Color.CYAN);
186         next.setBounds( x: 600, y: 690, width: 100, height: 30);
187         next.addActionListener( this);
188         add(next);
189
190
191
192         getContentPane().setBackground(new Color( r: 118, g: 224, b: 245));
193         setLayout(null);
194         setSize( width: 850, height: 800);
195         setLocation( x: 360, y: 40);
196         setVisible(true);
197
198     }
199
200     @Override
201     public void actionPerformed(ActionEvent e) {
202
203         String formno = first;
204         String name = textName.getText();
205
206     }
```

```
11 public class Signup extends JFrame implements ActionListener {
203     public void actionPerformed(ActionEvent e) {
207         String fname = textName.getText();
208         String dob = ((JTextField) dateChooser.getDateEditor().getUiComponent()).getText();
209         String gender = null;
210         if(r1.isSelected()){
211             gender = "MALE";
212         }else if(r2.isSelected()){
213             gender = "FEMALE";
214         }
215         String email = textEmail.getText();
216         String marital = null;
217         if(m1.isSelected()){
218             marital = "MARRIED";
219         }else if(m2.isSelected()){
220             marital = "UNMARRIED";
221         } else if (m3.isSelected()) {
222             marital = "OTHER";
223         }
224         String add = textAdd.getText();
225         String city = textCity.getText();
226         String pincode = textPincode.getText();
227         String state = textState.getText();
228
229         try{
230             if(textName.getText().equals("")){
231                 JOptionPane.showMessageDialog( parentComponent: null, message: "Fill all the details");
232             }else {
233                 Co co1 = new Co();
234                 String q = "insert into signup values('"+formno+"','"+name+"', '"+fname+"', '"+dob+"', '"+gender+"', '"+email+"', '"+marital+"', '"+ad
235                 co1.statement.executeUpdate(q);
236                 new Signup2(formno);
237             }
```

**


```
11 public class Signup extends JFrame implements ActionListener {
123     public void actionPerformed(ActionEvent e) {
124
125         String pincode = textPincode.getText();
126         String state = textState.getText();
127
128         try{
129             if(textName.getText().equals("")){
130                 JOptionPane.showMessageDialog(parentComponent: null, message: "Fill all the details");
131             }else {
132                 Co co1 = new Co();
133                 String q = "insert into signup values('"+formno+"','"+name+"', '"+fname+"', '"+dob+"', '"+gender+"','"+email+"','"+marital+"','"+ad
134                 co1.statement.executeUpdate(q);
135                 new Signup2(formno);
136                 setVisible(false);
137             }
138         }catch (Exception E){
139             E.printStackTrace();
140         }
141     }
142
143     public static void main(String[] args){ new Signup(); }
```

Signup2.java

```
1 package bank.management.system;
2
3 import javax.swing.*;
4 import java.awt.*;
5 import java.awt.event.ActionEvent;
6 import java.awt.event.ActionListener;
7
8 public class Signup2 extends JFrame implements ActionListener {
9     JComboBox combobox, combobox2, combobox3, combobox4, combobox5; 6 usages
10
11     JTextField textAadhar, textPan; 7 usages
12     JRadioButton r1, r2, e1, e2; 7 usages
13     JButton next; 7 usages
14
15     String formno; 3 usages
16
17     Signup2(String formno){ 2 usages
18         super( title: "APPLICATION FORM");
19
20         ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource( name: "icons/bank3.jpg"));
21         Image i2 = i1.getImage().getScaledInstance( width: 100, height: 100, Image.SCALE_DEFAULT);
22         ImageIcon i3 = new ImageIcon(i2);
23         JLabel image = new JLabel(i3);
24         image.setBounds( x: 150, y: 5, width: 100, height: 100);
25         add(image);
26
27         this.formno = formno;
28
29         JLabel l1 = new JLabel( text: "Page 2");
30         l1.setBounds( x: 300, y: 30, width: 600, height: 40);
31         l1.setFont(new Font( name: "Raleway",Font.BOLD, size: 22));
32         add(l1);
```

**

```
Bank Management System Version control Login 19 40 Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java

8 public class Signup2 extends JFrame implements ActionListener {
17 Signup2(String formno){ 2 usages
33
34 JLabel l2 = new JLabel( text: "Additional Details");
35 l2.setBounds( x: 300, y: 60, width: 600, height: 40);
36 l2.setFont(new Font( name: "Raleway",Font.BOLD, size: 22));
37 add(l2);
38
39 JLabel l3 = new JLabel( text: "RELIGION :");
40 l3.setBounds( x: 100, y: 120, width: 100, height: 30);
41 l3.setFont(new Font( name: "Raleway",Font.BOLD, size: 18));
42 add(l3);
43
44 String religion[] = {"Hinduism","Muslim","Sikhism","christian","Other"};
45 combobox = new JComboBox(religion);
46 combobox.setBounds( x: 350, y: 120, width: 320, height: 30);
47 combobox.setFont(new Font( name: "Raleway",Font.BOLD, size: 14));
48 combobox.setBackground(new Color( r: 148, g: 203, b: 242));
49 add(combobox);
50
51 JLabel l4 = new JLabel( text: "CATEGORY :");
52 l4.setBounds( x: 100, y: 170, width: 150, height: 30);
53 l4.setFont(new Font( name: "Raleway",Font.BOLD, size: 18));
54 add(l4);
55
56 String category[] = {"General","OBC","SC","ST","Other"};
57 combobox2 = new JComboBox(category);
58 combobox2.setBounds( x: 350, y: 170, width: 320, height: 30);
59 combobox2.setFont(new Font( name: "Raleway",Font.BOLD, size: 14));
60 combobox2.setBackground(new Color( r: 148, g: 203, b: 242));
61 add(combobox2);
62

Bank Management System > src > bank > management > system > Signup2 > e2 12:33 CRLF UTF-8 4 spaces
```

```
Bank Management System Version control Login 19 40 Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java

8 public class Signup2 extends JFrame implements ActionListener {
17 Signup2(String formno){ 2 usages
63
64 JLabel l5 = new JLabel( text: "INCOME :");
65 l5.setBounds( x: 100, y: 220, width: 150, height: 30);
66 l5.setFont(new Font( name: "Raleway",Font.BOLD, size: 18));
67 add(l5);
68
69 String income[] = {"Null","<1,50,000","<2,50,000","5,00,000","Upto 10,00,000","Above 10,00,000"};
70 combobox3 = new JComboBox(income);
71 combobox3.setBounds( x: 350, y: 220, width: 320, height: 30);
72 combobox3.setFont(new Font( name: "Raleway",Font.BOLD, size: 14));
73 combobox3.setBackground(new Color( r: 148, g: 203, b: 242));
74 add(combobox3);
75
76 JLabel l6 = new JLabel( text: "EDUCATIONAL :");
77 l6.setBounds( x: 100, y: 270, width: 170, height: 30);
78 l6.setFont(new Font( name: "Raleway",Font.BOLD, size: 18));
79 add(l6);
80
81 String educational[] = {"Non-Graduate","Graduate","Post-Graduate","Doctrate","Other"};
82 combobox4 = new JComboBox(educational);
83 combobox4.setBounds( x: 350, y: 270, width: 320, height: 30);
84 combobox4.setFont(new Font( name: "Raleway",Font.BOLD, size: 14));
85 combobox4.setBackground(new Color( r: 148, g: 203, b: 242));
86 add(combobox4);
87
88 JLabel l7 = new JLabel( text: "OCCUPATION :");
89 l7.setBounds( x: 100, y: 320, width: 170, height: 30);
90 l7.setFont(new Font( name: "Raleway",Font.BOLD, size: 18));
91 add(l7);
92
93 String occupation[] = {"Salaried","Self-Employed","Business","Student","Retired","Other"};

Bank Management System > src > bank > management > system > Signup2 > e2 12:33 CRLF UTF-8 4 spaces
```

**

```
Bank Management System Version control Login 19 40 x
Wit.java Fastc.java Signup2.java x Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java
8 public class Signup2 extends JFrame implements ActionListener {
17 Signup2(String formno){ 2 usages
93 JComboBox comboBox5 = new JComboBox(occupation);
94 comboBox5.setBounds( x: 350, y: 320, width: 320, height: 30);
95 comboBox5.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
96 comboBox5.setBackground(new Color( r: 148, g: 203, b: 242));
97 add(comboBox5);
98
99 JLabel l8 = new JLabel( text: "AADHAR NUMBER :");
100 l8.setBounds( x: 100, y: 370, width: 200, height: 30);
101 l8.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 18));
102 add(l8);
103
104 JTextField textAadhar = new JTextField();
105 textAadhar.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
106 textAadhar.setBounds( x: 350, y: 370, width: 320, height: 30);
107 textAadhar.setBackground(new Color( r: 148, g: 203, b: 242));
108 add(textAadhar);
109
110
111 JLabel l9 = new JLabel( text: "PAN NUMBER :");
112 l9.setBounds( x: 100, y: 420, width: 170, height: 30);
113 l9.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 18));
114 add(l9);
115
116 JTextField textPan = new JTextField();
117 textPan.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
118 textPan.setBounds( x: 350, y: 420, width: 320, height: 30);
119 textPan.setBackground(new Color( r: 148, g: 203, b: 242));
120 add(textPan);
121
122 JLabel l10 = new JLabel( text: "SENIOR CITIZEN :");
```

Bank Management System > src > bank > management > system > Signup2 > e2 12:33 CRLF UTF-8 4 spaces

```
Bank Management System Version control Login 19 40 x
Wit.java Fastc.java Signup2.java x Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java
8 public class Signup2 extends JFrame implements ActionListener {
17 Signup2(String formno){ 2 usages
123 l10.setBounds( x: 100, y: 470, width: 200, height: 30);
124 l10.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 18));
125 add(l10);
126
127 JRadioButton r1 = new JRadioButton( text: "YES");
128 r1.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
129 r1.setBackground(new Color( r: 148, g: 203, b: 242));
130 r1.setBounds( x: 350, y: 470, width: 70, height: 30);
131 add(r1);
132
133 JRadioButton r2 = new JRadioButton( text: "NO");
134 r2.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
135 r2.setBackground(new Color( r: 148, g: 203, b: 242));
136 r2.setBounds( x: 500, y: 470, width: 70, height: 30);
137 add(r2);
138
139 ButtonGroup buttonGroup = new ButtonGroup();
140 buttonGroup.add(r1);
141 buttonGroup.add(r2);
142
143 JLabel l11 = new JLabel( text: "EXISTING ACCOUNT :");
144 l11.setBounds( x: 100, y: 520, width: 200, height: 30);
145 l11.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 18));
146 add(l11);
147
148 JRadioButton e1 = new JRadioButton( text: "YES");
149 e1.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
150 e1.setBackground(new Color( r: 148, g: 203, b: 242));
151 e1.setBounds( x: 350, y: 520, width: 70, height: 30);
152 add(e1);
```

Bank Management System > src > bank > management > system > Signup2 > e2 12:33 CRLF UTF-8 4 spaces

**

```
Bank Management System Version control Login 19 40 12:33 CRLF UTF-8 4 spaces

Wit.java Fastc.java Signup2.java x Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java

8 public class Signup2 extends JFrame implements ActionListener {
17 Signup2(String formno){ 2 usages

153
154     e2 = new JRadioButton( text: "NO");
155     e2.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
156     e2.setBackground(new Color( r: 148, g: 203, b: 242));
157     e2.setBounds( x: 500, y: 520, width: 70, height: 30);
158     add(e2);
159
160     ButtonGroup buttonGroup1 = new ButtonGroup();
161     buttonGroup1.add(e1);
162     buttonGroup1.add(e2);
163
164     JLabel l12 = new JLabel( text: "FORM No. :");
165     l12.setBounds( x: 650, y: 10, width: 100, height: 30);
166     l12.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 18));
167     add(l12);
168
169     JLabel l13 = new JLabel(formno);
170     l13.setBounds( x: 760, y: 10, width: 60, height: 30);
171     l13.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 18));
172     add(l13);
173
174     next = new JButton( text: "NEXT");
175     next.setFont(new Font( name: "RaLeWay", Font.BOLD, size: 18));
176     next.setBackground(Color.BLACK);
177     next.setForeground(Color.CYAN);
178     next.setBounds( x: 570, y: 640, width: 100, height: 30);
179     next.addActionListener( l: this);
180     add(next);
181
182

Bank Management System > src > bank > management > system > Signup2 > e2 12:33 CRLF UTF-8 4 spaces
```

```
Bank Management System Version control Login 19 40 12:33 CRLF UTF-8 4 spaces

Wit.java Fastc.java Signup2.java x Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java

8 public class Signup2 extends JFrame implements ActionListener {
17 Signup2(String formno){ 2 usages

185     setLayout(null);
186     setSize( width: 850, height: 750);
187     setLocation( x: 450, y: 80);
188     getContentPane().setBackground(new Color( r: 108, g: 182, b: 235));
189     setVisible(true);
190 }
191
192 @Override
193 public void actionPerformed(ActionEvent e) {
194     String rel = (String) combobox.getSelectedItemAt();
195     String cate = (String) combobox2.getSelectedItemAt();
196     String inc = (String) combobox3.getSelectedItemAt();
197     String edu = (String) combobox4.getSelectedItemAt();
198     String occ = (String) combobox5.getSelectedItemAt();
199
200     String pan = textPan.getText();
201     String aadhar = textAadhar.getText();
202
203     String scitizen = " ";
204     if ((r1.isSelected())){
205         scitizen = "Yes";
206     }else if (r2.isSelected()){
207         scitizen = "No";
208     }
209
210     String eaccount = " ";
211     if ((e1.isSelected())){
212         eaccount = "Yes";
213     }else if (e2.isSelected()){
214         eaccount = "No";
215     }
216 }

Bank Management System > src > bank > management > system > Signup2 > e2 12:33 CRLF UTF-8 4 spaces
```

```
Bank Management System  Version control  Login  19 40 16:33 CRLF UTF-8 4 spaces

Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java

8 public class Signup2 extends JFrame implements ActionListener {
193 public void actionPerformed(ActionEvent e) {
212     eaccount = "Yes";
213 }else if (e2.isSelected()){
214     eaccount = "No";
215 }
216 try{
217     if(textPan.getText().equals("") || textAadhar.getText().equals("")){
218         JOptionPane.showMessageDialog( parentComponent: null, message: "Fill all the fields");
219     }else {
220         Co c1 = new Co();
221         String q = "insert into Signuptwo values('"+formno+"', '"+rel+"', '"+cate+"', '"+inc+"', '"+edu+"', '"+occ+"', '"+pan+"', '"+aadhar+"', '"+scitizen+'', '"+eacco";
222         c1.statement.executeUpdate(q);
223         new Signup3(formno);
224         setVisible(false);
225     }
226 }catch(Exception E){
227     E.printStackTrace();
228 }
229 }
230
231 }
232
233 public static void main(String[] args){ new Signup2( formno: ""); }
236 }
237

Bank Management System > src > bank > management > system > Signup2 > e2
```

Signup3.java

```
Bank Management System  Version control  Login  3 35 13:41 CRLF UTF-8 4 spaces

Proje  Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java

Bank Management Sy
  > idea
  > out
  > src
    > bank.managemen
      > Balen
      > Co
      > Deposit
      > Fastc
      > Login
      > main_Class
      > Minis
      > Pinc
      > Signup
      > Signup2
      > Signup3
      > Wit
    > icons
    > .gitignore
    > Bank Management S
  > External Libraries
    > C:\Program F
    > jcalendar-tz-1.3.3-4
    > mysql-connector-jav
    > Scratches and Console

1 package bank.management.system;
2
3 import javax.swing.*;
4 import java.awt.*;
5 import java.awt.event.ActionEvent;
6 import java.awt.event.ActionListener;
7 import java.util.Random;
8
9 public class Signup3 extends JFrame implements ActionListener {
10
11     JRadioButton r1, r2, r3, r4; 7 usages
12     JCheckBox c1, c2, c3, c4, c5, c6; 6 usages
13     JButton s, c; 8 usages
14     String formno; 3 usages
15     Signup3(String formno){ 2 usages
16
17         this.formno = formno;
18
19         ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource( name: "icons/bank3.jpg"));
20         Image i2 = i1.getImage().getScaledInstance( width: 100, height: 100, Image.SCALE_DEFAULT);
21         ImageIcon i3 = new ImageIcon(i2);
22         JLabel image = new JLabel(i3);
23         image.setBounds( x: 150, y: 5, width: 100, height: 100);
24         add(image);
25
26         JLabel l1 = new JLabel( text: "Page 3");
27         l1.setBounds( x: 280, y: 40, width: 400, height: 40);
28         l1.setFont(new Font( name: "Raleway", Font.BOLD, size: 22));
29         add(l1);
30
31         JLabel l2 = new JLabel( text: "Account Details");
32         l2.setBounds( x: 280, y: 70, width: 400, height: 40);
```

**

```
Bank Management System Version control Login 35 4 spaces

Project Bank Management System
  .idea
  out
  src
    bank.management
      Balen
      Co
      Deposit
      Fastc
      Login
      main_Class
      Minis
      Pinc
      Signup
      Signup2
      Signup3
      Wit
    icons
    .gitignore
    Bank Management System
  External Libraries
    < 23 > C:\Program F
    jcalendar-tz-1.3.3-4
    mysql-connector-jav
  Scratches and Console

9 public class Signup3 extends JFrame implements ActionListener {
15 Signup3(String formno){ 2 usages
33 l2.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 22));
34 add(l2);
35
36 JLabel l3 = new JLabel( text: "ACCOUNT TYPE :");
37 l3.setBounds( x: 100, y: 140, width: 200, height: 30);
38 l3.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 20));
39 add(l3);
40
41 r1 = new JRadioButton( text: "SAVING ACCOUNT");
42 r1.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
43 r1.setBackground(new Color( r: 148, g: 203, b: 242));
44 r1.setBounds( x: 100, y: 180, width: 200, height: 30);
45 add(r1);
46
47 r2 = new JRadioButton( text: "FIXED DEPOSIT ACCOUNT");
48 r2.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
49 r2.setBackground(new Color( r: 148, g: 203, b: 242));
50 r2.setBounds( x: 350, y: 180, width: 300, height: 30);
51 add(r2);
52
53 r3 = new JRadioButton( text: "CURRENT ACCOUNT");
54 r3.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
55 r3.setBackground(new Color( r: 148, g: 203, b: 242));
56 r3.setBounds( x: 100, y: 220, width: 200, height: 30);
57 add(r3);
58
59 r4 = new JRadioButton( text: "RECURRING DEPOSIT ACCOUNT");
60 r4.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 14));
61 r4.setBackground(new Color( r: 148, g: 203, b: 242));
62 r4.setBounds( x: 350, y: 220, width: 300, height: 30);

Bank Management System > src > bank > management > system > main_Class 134:1 CRLF UTF-8 4 spaces
```

```
Bank Management System Version control Login 35 4 spaces

Project Bank Management System
  .idea
  out
  src
    bank.management
      Balen
      Co
      Deposit
      Fastc
      Login
      main_Class
      Minis
      Pinc
      Signup
      Signup2
      Signup3
      Wit
    icons
    .gitignore
    Bank Management System
  External Libraries
    < 23 > C:\Program F
    jcalendar-tz-1.3.3-4
    mysql-connector-jav
  Scratches and Console

9 public class Signup3 extends JFrame implements ActionListener {
15 Signup3(String formno){ 2 usages
63 add(r4);
64
65 ButtonGroup buttonGroup = new ButtonGroup();
66 buttonGroup.add(r1);
67 buttonGroup.add(r2);
68 buttonGroup.add(r3);
69 buttonGroup.add(r4);
70
71 JLabel l4 = new JLabel( text: "CARD NUMBER :");
72 l4.setBounds( x: 100, y: 300, width: 200, height: 30);
73 l4.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 20));
74 add(l4);
75
76 JLabel l5 = new JLabel( text: "(Your 16-Digit Card Number)");
77 l5.setBounds( x: 100, y: 330, width: 400, height: 20);
78 l5.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 12));
79 add(l5);
80
81 JLabel l6 = new JLabel( text: "XXXX-XXXX-XXXX-4841");
82 l6.setBounds( x: 350, y: 300, width: 400, height: 30);
83 l6.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 20));
84 add(l6);
85
86 JLabel l7 = new JLabel( text: "(It would appear on atm card/cheque book and statements)");
87 l7.setBounds( x: 350, y: 330, width: 600, height: 30);
88 l7.setFont(new Font( name: "RaLeWay",Font.BOLD, size: 12));
89 add(l7);
90
91 JLabel l8 = new JLabel( text: "PIN :");
92 l8.setBounds( x: 100, y: 380, width: 100, height: 30);

Bank Management System > src > bank > management > system > main_Class 134:1 CRLF UTF-8 4 spaces
```

**


```
Bank Management System  Version control  Login  35  3  35  ^  v  @

Project  Wit.java  Fastc.java  Signup2.java  Signup3.java  Signup.java  Deposit.java  Pinc.java  Balen.java  Login.java

Bank Management System
  > .idea
  > out
  > src
    > bank.management
      Balen
      Co
      Deposit
      Fastc
      Login
      main_Class
      Minis
      Pinc
      Signup
      Signup2
      Signup3
      Wit
    > icons
    > .gitignore
    > Bank Management System
  > External Libraries
    > < 23 > C:\Program F
    > jcalendar-tz-1.3.3-4
    > mysql-connector-jav
  > Scratches and Console:

9  public class Signup3 extends JFrame implements ActionListener {
192  public void actionPerformed(ActionEvent e) {
213      fac += "ATM CARD";
214  }else if(c2.isSelected()){
215      fac += "INTERNET BANKING";
216  }else if(c3.isSelected()){
217      fac += "MOBILE BANKING";
218  }else if(c4.isSelected()){
219      fac += "EMAIL ALERTS";
220  }else if(c5.isSelected()){
221      fac += "CHEQUE BOOK";
222  }else if(c6.isSelected()){
223      fac += "E-STATEMENT";
224  }
225
226  try{
227      if(e.getSource()==s){
228          if(atype.equals("")){
229              JOptionPane.showMessageDialog( parentComponent: null, message: "Fill all the fields");
230          }else{
231              Co c1 = new Co();
232              String q1 = "insert into signupthree values('"+formno+"','"+atype+"','"+cardno+"','"+pin+"','"+fac+"')";
233              String q2 = "insert into login values('"+formno+"','"+cardno+"','"+pin+"')";
234              c1.statement.executeUpdate(q1);
235              c1.statement.executeUpdate(q2);
236              JOptionPane.showMessageDialog( parentComponent: null, message: "CARD NUMBER :"+cardno+"\n PIN :"+pin );
237              new Deposit(pin);
238              setVisible(false);
239          }
240      }else if (e.getSource()==c){
241          System.exit( status: 0);
242      }
243  }
244  }

Bank Management System > src > bank > management > system > main_Class 134:1 CRLF UTF-8 4 spaces
```

```
Bank Management System  Version control  Login  35  3  35  ^  v  @

Project  Wit.java  Fastc.java  Signup2.java  Signup3.java  Signup.java  Deposit.java  Pinc.java  Balen.java  Login.java

Bank Management System
  > .idea
  > out
  > src
    > bank.management
      Balen
      Co
      Deposit
      Fastc
      Login
      main_Class
      Minis
      Pinc
      Signup
      Signup2
      Signup3
      Wit
    > icons
    > .gitignore
    > Bank Management System
  > External Libraries
    > < 23 > C:\Program F
    > jcalendar-tz-1.3.3-4
    > mysql-connector-jav
  > Scratches and Console:

9  public class Signup3 extends JFrame implements ActionListener {
192  public void actionPerformed(ActionEvent e) {
229      JOptionPane.showMessageDialog( parentComponent: null, message: "Fill all the fields");
230  }else{
231      Co c1 = new Co();
232      String q1 = "insert into signupthree values('"+formno+"','"+atype+"','"+cardno+"','"+pin+"','"+fac+"')";
233      String q2 = "insert into login values('"+formno+"','"+cardno+"','"+pin+"')";
234      c1.statement.executeUpdate(q1);
235      c1.statement.executeUpdate(q2);
236      JOptionPane.showMessageDialog( parentComponent: null, message: "CARD NUMBER :"+cardno+"\n PIN :"+pin );
237      new Deposit(pin);
238      setVisible(false);
239  }
240  }else if (e.getSource()==c){
241      System.exit( status: 0);
242  }
243  }
244  }
245  }catch(Exception E){
246      E.printStackTrace();
247  }
248  }
249
250  public static void main(String[] args){ new Signup3( formno: " "); }
251  }
252
253
254

Bank Management System > src > bank > management > system > main_Class 134:1 CRLF UTF-8 4 spaces
```

**

Deposit.java

```
Bank Management System  Version control  Login  Run  Debug  Search  Settings  Window  Help  x

Wit.java  Fastc.java  Signup2.java  Signup3.java  Signup.java  Deposit.java x  Pinc.java  Balen.java  Login.java  Minis.java

1  package bank.management.system;
2
3  import javax.swing.*;
4  import java.awt.*;
5  import java.awt.event.ActionEvent;
6  import java.awt.event.ActionListener;
7  import java.util.Date;
8
9  public class Deposit extends JFrame implements ActionListener {
10      String pin; 4 usages
11      TextField textField; 7 usages
12
13      JButton b1, b2; 8 usages
14
15      Deposit(String pin){ 3 usages
16          this.pin = pin;
17
18          ImageIcon i1 = new ImageIcon(ClassLoader.getResource("icons/atm.jpg"));
19          Image i2 = i1.getImage().getScaledInstance(1550, 830, Image.SCALE_DEFAULT);
20          ImageIcon i3 = new ImageIcon(i2);
21          JLabel l3 = new JLabel(i3);
22          l3.setBounds(x: 0, y: 0, width: 1550, height: 830);
23          add(l3);
24
25          JLabel label1 = new JLabel(text: "ENTER AMOUNT YOU WANT TO DEPOSIT");
26          label1.setFont(new Font(name: "System", Font.ITALIC, size: 16));
27          label1.setBounds(x: 500, y: 420, width: 400, height: 35);
28          l3.add(label1);
29
30          textField = new TextField();
31          textField.setBounds(x: 500, y: 470, width: 320, height: 25);
32          textField.setFont(new Font(name: "RaLeway", Font.ITALIC, size: 22));
33      }
34
35      @Override
36      public void actionPerformed(ActionEvent e) {
37          // TODO: Implement actionPerformed method
38      }
39  }

Bank Management System > src > bank > management > system > Deposit 6:38 CRLF UTF-8 4 spaces
```

```
Bank Management System  Version control  Login  Run  Debug  Search  Settings  Window  Help  x

Wit.java  Fastc.java  Signup2.java  Signup3.java  Signup.java  Deposit.java x  Pinc.java  Balen.java  Login.java  Minis.java

9  public class Deposit extends JFrame implements ActionListener {
15      Deposit(String pin){ 3 usages
33      textField.setBackground(new Color(r: 148, g: 203, b: 242));
34      l3.add(textField);
35
36      b1 = new JButton(text: "DEPOSIT");
37      b1.setFont(new Font(name: "RaLeway", Font.ITALIC, size: 14));
38      b1.setBackground(new Color(r: 148, g: 203, b: 242));
39      b1.setForeground(Color.BLACK);
40      b1.setBounds(x: 500, y: 520, width: 120, height: 30);
41      b1.addActionListener(this);
42      l3.add(b1);
43
44      b2 = new JButton(text: "BACK");
45      b2.setFont(new Font(name: "RaLeway", Font.ITALIC, size: 14));
46      b2.setBackground(new Color(r: 148, g: 203, b: 242));
47      b2.setForeground(Color.BLACK);
48      b2.setBounds(x: 700, y: 520, width: 120, height: 30);
49      b2.addActionListener(this);
50      l3.add(b2);
51
52      setLayout(null);
53      setSize(width: 1550, height: 1000);
54      setLocation(x: 0, y: 0);
55      setVisible(true);
56  }
57
58  @Override
59  public void actionPerformed(ActionEvent e) {
60      try {
61          // TODO: Implement actionPerformed method
62      }
63  }

Bank Management System > src > bank > management > system > Deposit 6:38 CRLF UTF-8 4 spaces
```

**

```
Bank Management System Version control
Login
Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java
9 public class Deposit extends JFrame implements ActionListener {
61 public void actionPerformed(ActionEvent e) {
62     try {
63         String amount = textField.getText();
64         Date date = new Date();
65         if(e.getSource() == b1){
66             if(textField.getText().equals("")){
67                 JOptionPane.showMessageDialog( parentComponent: null, message: "Please enter the Amount you want to Deposit");
68             }else {
69                 Co c = new Co();
70                 c.statement.executeUpdate( sql: "insert into bank values('"+pin+"','"+date+"','Deposit', '"+amount+"')");
71                 JOptionPane.showMessageDialog( parentComponent: null, message: "Rs. "+amount+"Deposited Successfully");
72                 setVisible(false);
73                 new main_Class(pin);
74             }
75         }else if(e.getSource() == b2){
76             setVisible(false);
77             new main_Class(pin);
78         }
79     }catch(Exception E){
80         E.printStackTrace();
81     }
82 }
83
84 public static void main(String[] args) { new Deposit( pin: " "); }
87 }
88

Bank Management System > src > bank > management > system > Deposit 6:38 CRLF UTF-8 4 spaces
```

Pinc.java

```
Bank Management System Version control
Login
Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java
Project
Bank Management System
  .idea
  out
  src
    bank.managemen
      Balen
      Co
      Deposit
      Fastc
      Login
      main_Class
      Minis
      Pinc
      Signup
      Signup2
      Signup3
      Wit
    icons
  .gitignore
  Bank Management S
External Libraries
  C:\Program F
  jcalendar-tz-1.3.3-4
  mysql-connector-jav
Scratches and Console:
1 package bank.management.system;
2
3 import javax.swing.*;
4 import java.awt.*;
5 import java.awt.event.ActionEvent;
6 import java.awt.event.ActionListener;
7
8 public class Pinc extends JFrame implements ActionListener {
9
10     JButton b1, b2; 8 usages
11     JPasswordField p1, p2; 7 usages
12     String pin; 6 usages
13     Pinc(String pin){ 2 usages
14
15         this.pin = pin;
16
17
18         ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource( name: "icons/atm.jpg"));
19         Image i2 = i1.getImage().getScaledInstance( width: 1550, height: 830, Image.SCALE_DEFAULT);
20         ImageIcon i3 = new ImageIcon(i2);
21         JLabel l3 = new JLabel(i3);
22         l3.setBounds( x: 0, y: 0, width: 1550, height: 830);
23         add(l3);
24
25         JLabel label1 = new JLabel( text: "CHANGE YOUR PIN");
26         label1.setFont(new Font( name: "System", Font.ITALIC, size: 18));
27         label1.setBounds( x: 500, y: 400, width: 400, height: 35);
28         l3.add(label1);
29
30         JLabel label2 = new JLabel( text: "NEW PIN :");
31         label2.setFont(new Font( name: "System", Font.ITALIC, size: 14));
32         label2.setBounds( x: 500, y: 440, width: 400, height: 35);
33     }
34 }

Bank Management System > src > bank > management > system > Pinc > Pinc 6:11 CRLF UTF-8 4 spaces
```

**

```
8 public class Pinc extends JFrame implements ActionListener {
13     Pinc(String pin){ 2 usages
33         l3.add(label2);
34
35         p1 = new JPasswordField();
36         p1.setBounds( x: 670, y: 445, width: 200, height: 25);
37         p1.setFont(new Font( name: "RaLeWay", Font.ITALIC, size: 22));
38         p1.setBackground(new Color( r: 148, g: 203, b: 242));
39         l3.add(p1);
40
41         JLabel label3 = new JLabel( text: "RE-ENTER NEW PIN :");
42         label3.setFont(new Font( name: "System",Font.ITALIC, size: 14));
43         label3.setBounds( x: 500, y: 470, width: 400, height: 35);
44         l3.add(label3);
45
46         p2 = new JPasswordField();
47         p2.setBounds( x: 670, y: 475, width: 200, height: 25);
48         p2.setFont(new Font( name: "RaLeWay", Font.ITALIC, size: 22));
49         p2.setBackground(new Color( r: 148, g: 203, b: 242));
50         l3.add(p2);
51
52
53         b1 = new JButton( text: "CHANGE");
54         b1.setFont(new Font( name: "RaLeWay",Font.ITALIC, size: 14));
55         b1.setBackground(new Color( r: 148, g: 203, b: 242));
56         b1.setForeground(Color.BLACK);
57         b1.setBounds( x: 500, y: 520, width: 120, height: 30);
58         b1.addActionListener( l: this);
59         l3.add(b1);
60
61
62 }
```

```
63         b2 = new JButton( text: "BACK");
64         b2.setFont(new Font( name: "RaLeWay",Font.ITALIC, size: 14));
65         b2.setBackground(new Color( r: 148, g: 203, b: 242));
66         b2.setForeground(Color.BLACK);
67         b2.setBounds( x: 700, y: 520, width: 120, height: 30);
68         b2.addActionListener( l: this);
69         l3.add(b2);
70
71
72         setLayout(null);
73         setSize( width: 1550, height: 1000);
74         setLocation( x: 0, y: 0);
75         setVisible(true);
76
77
78     }
79
80
81     @Override
82     public void actionPerformed(ActionEvent e) {
83
84         try{
85
86             String pin1 = p1.getText();
87             String pin2 = p2.getText();
88
89             if(!pin1.equals(pin2)){
90
91                 JOptionPane.showMessageDialog( parentComponent: null, message: "Entered PIN does not match");
92
93             }
```

```
8 public class Pinc extends JFrame implements ActionListener {
84 public void actionPerformed(ActionEvent e) {
93     return;
94 }
95 if(e.getSource() == b1){
96     if(p1.getText().equals("")){
97         JOptionPane.showMessageDialog( parentComponent: null, message: "Enter New PIN");
98         return;
99     }
100 }
101 if(p2.getText().equals("")){
102     JOptionPane.showMessageDialog( parentComponent: null, message: "Re-Enter New PIN");
103     return;
104 }
105 }
106 Co c = new Co();
107 String q1 = "update bank set pin = '"+pin1+"' where pin='"+pin+"'";
108 String q2 = "update login set pin = '"+pin1+"' where pin='"+pin+"'";
109 String q3 = "update signupthree set pin = '"+pin1+"' where pin='"+pin+"'";
110
111 c.statement.executeUpdate(q1);
112 c.statement.executeUpdate(q2);
113 c.statement.executeUpdate(q3);
114
115 JOptionPane.showMessageDialog( parentComponent: null, message: "PIN changed successfully");
116 setVisible(false);
117 new main_Class(pin);
118
119 }else if(e.getSource() == b2){
120     new main_Class(pin);
121 }
122 }
```

```
8 public class Pinc extends JFrame implements ActionListener {
84     public void actionPerformed(ActionEvent e) {
111
112         c.statement.executeUpdate(q1);
113         c.statement.executeUpdate(q2);
114         c.statement.executeUpdate(q3);
115
116         JOptionPane.showMessageDialog( parentComponent: null, message: "PIN changed successfully");
117         setVisible(false);
118         new main_Class(pin);
119
120     }else if(e.getSource() == b2){
121         new main_Class(pin);
122         setVisible(false);
123     }
124 }catch(Exception E){
125     E.printStackTrace();
126 }
127 }
128
129 }
130
131 public static void main(String[] args) {
132     new Pinc("");
133 }
134 }
135 }
```

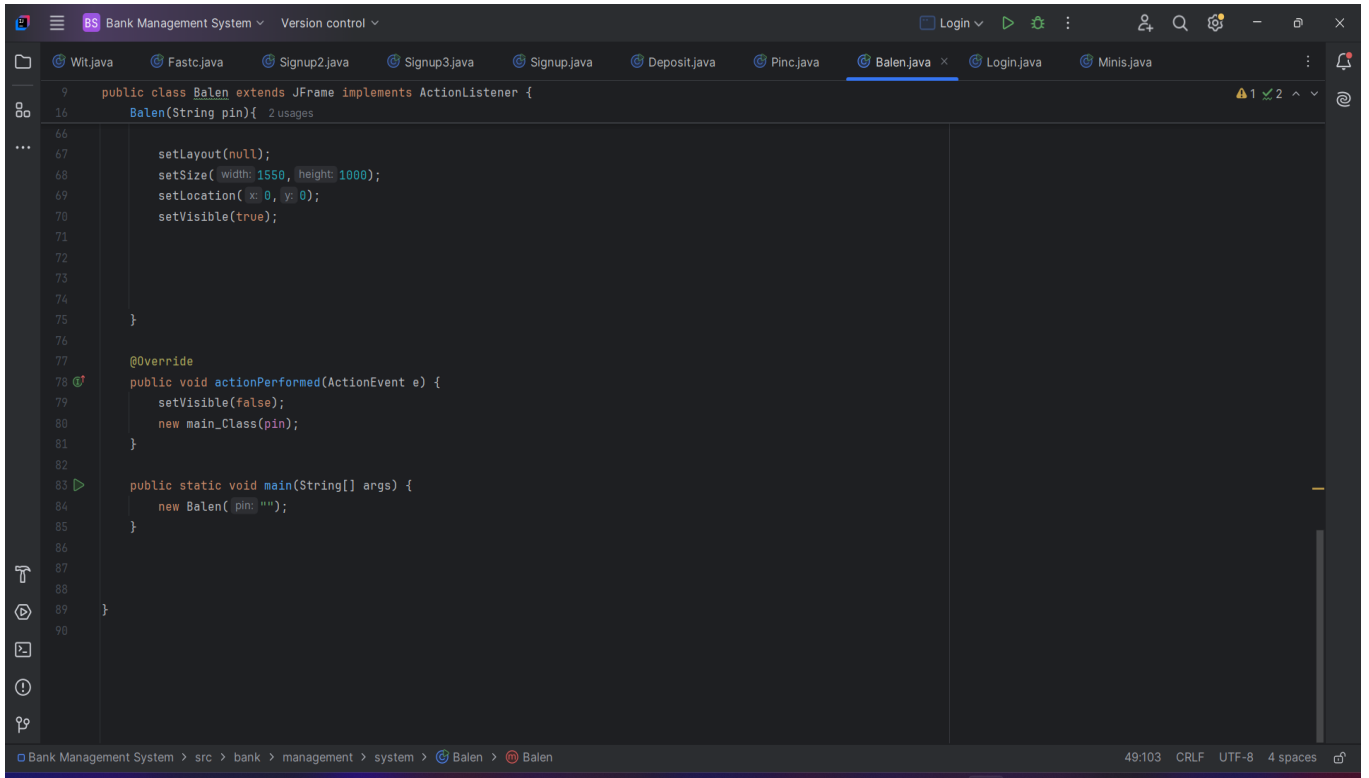
**

Balen.java

```
Bank Management System Version control
Login
Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java x Login.java Minis.java
1 package bank.management.system;
2
3 import javax.swing.*;
4 import java.awt.*;
5 import java.awt.event.ActionEvent;
6 import java.awt.event.ActionListener;
7 import java.sql.ResultSet;
8
9 public class Balen extends JFrame implements ActionListener {
10
11     String pin; 2 usages
12
13     JLabel label2; 5 usages
14     JButton b1; 7 usages
15
16     Balen(String pin){ 2 usages
17
18         this.pin = pin;
19
20
21         ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource("icons/atm.jpg"));
22         Image i2 = i1.getImage().getScaledInstance(1550, 830, Image.SCALE_DEFAULT);
23         ImageIcon i3 = new ImageIcon(i2);
24         JLabel l3 = new JLabel(i3);
25         l3.setBounds(0, 0, 1550, 830);
26         add(l3);
27
28         JLabel label1 = new JLabel(text: "YOUR CURRENT BALANCE IS Rs.");
29         label1.setFont(new Font(name: "System", Font.ITALIC, size: 16));
30         label1.setBounds(500, 410, 400, 35);
31         l3.add(label1);
32
33     }
34 }
```

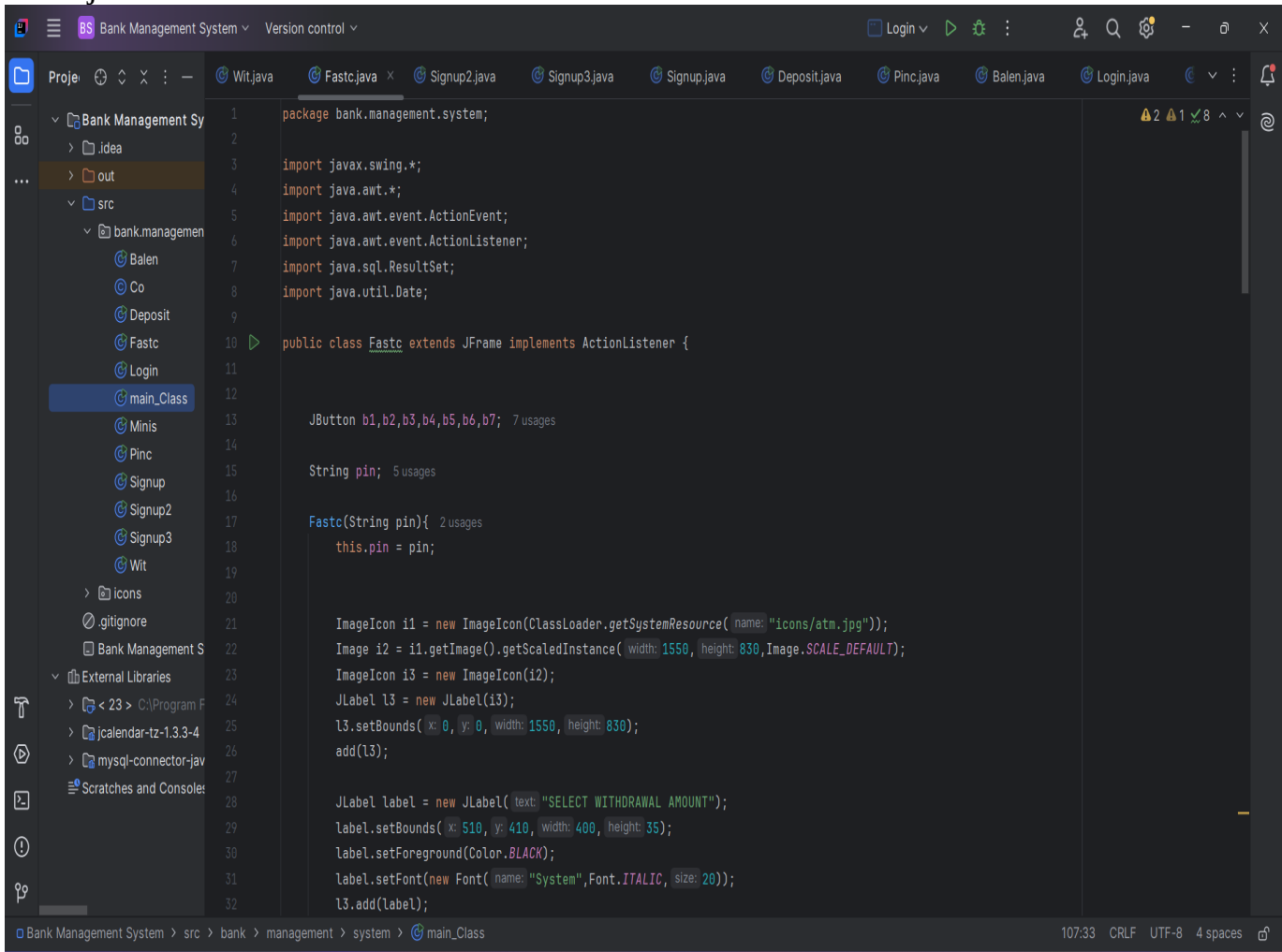
```
Bank Management System Version control
Login
Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java x Login.java Minis.java
9 public class Balen extends JFrame implements ActionListener {
16     Balen(String pin){ 2 usages
33         label2 = new JLabel();
34         label2.setFont(new Font(name: "System", Font.ITALIC, size: 16));
35         label2.setBounds(500, 440, 400, 35);
36         l3.add(label2);
37
38         b1 = new JButton(text: "BACK");
39         b1.setFont(new Font(name: "Raleway", Font.ITALIC, size: 14));
40         b1.setBackground(new Color(r: 148, g: 203, b: 242));
41         b1.setForeground(Color.BLACK);
42         b1.setBounds(500, 520, 120, 30);
43         b1.addActionListener(this);
44         l3.add(b1);
45
46         int balance = 0;
47         try{
48             Co c = new Co();
49             ResultSet resultSet = c.statement.executeQuery(sql: "Select * from bank where pin = '"+pin+"'");
50             while (resultSet.next()){
51                 if(resultSet.getString(columnLabel: "type").equals("Deposit")){
52                     balance += Integer.parseInt(resultSet.getString(columnLabel: "amount"));
53                 }else {
54                     balance -= Integer.parseInt(resultSet.getString(columnLabel: "amount"));
55                 }
56             }
57         }catch(Exception e){
58             e.printStackTrace();
59         }
60
61         label2.setText(""+balance);
62     }
```

**



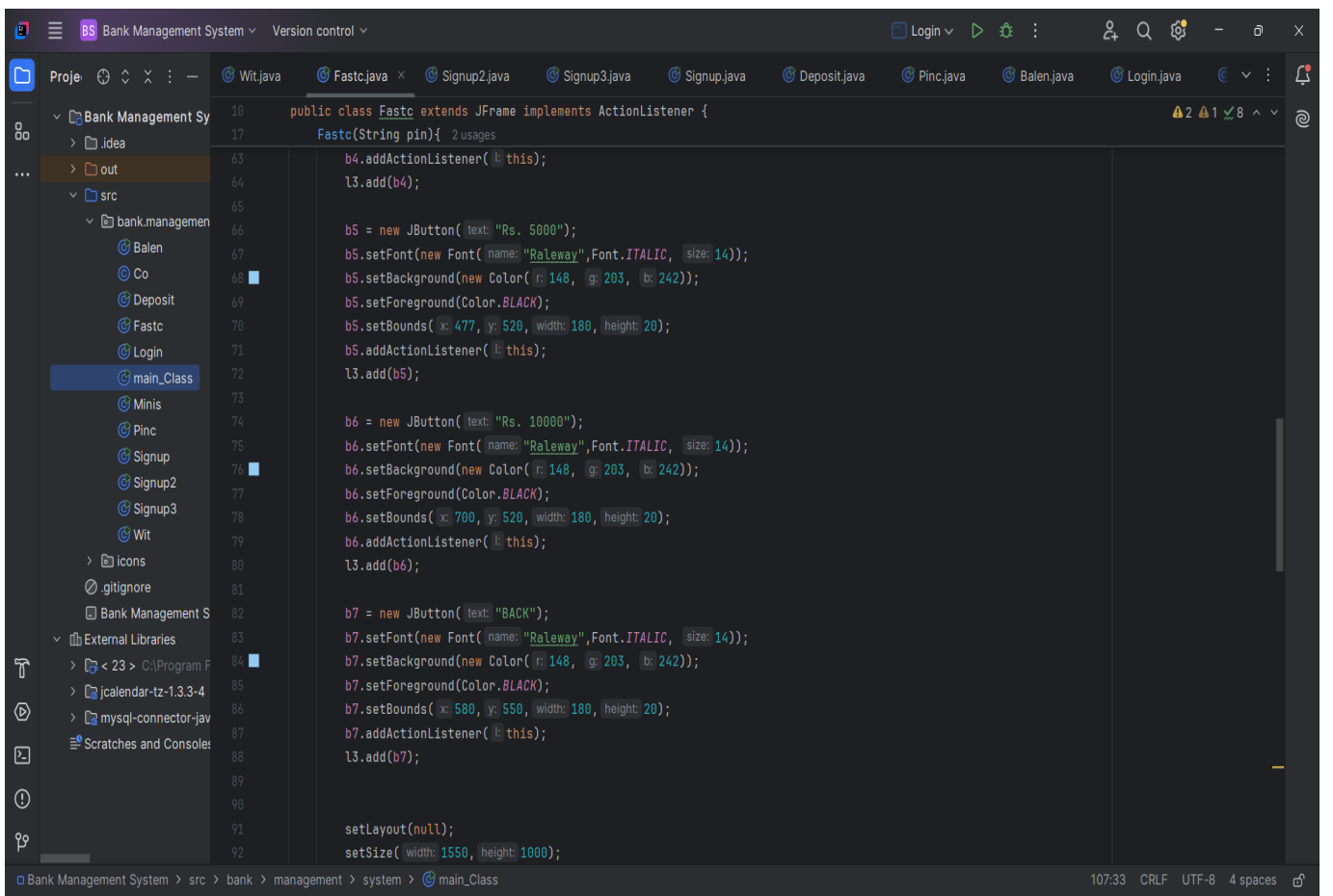
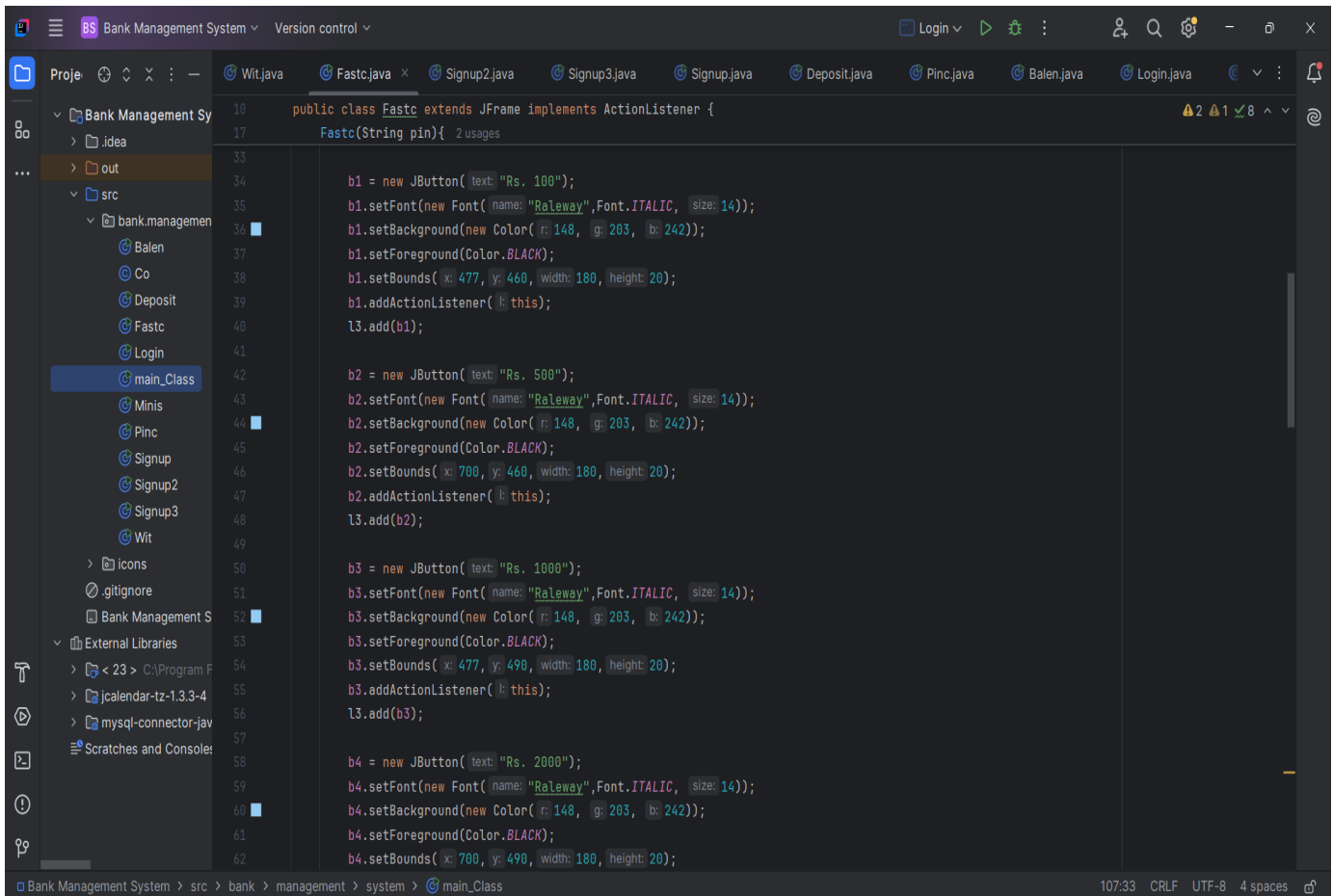
```
9 public class Balen extends JFrame implements ActionListener {
16     Balen(String pin){ 2 usages
66
67         setLayout(null);
68         setSize( width: 1550, height: 1000);
69         setLocation( x: 0, y: 0);
70         setVisible(true);
71
72
73
74     }
75
76
77     @Override
78     public void actionPerformed(ActionEvent e) {
79         setVisible(false);
80         new main_Class(pin);
81     }
82
83     public static void main(String[] args) {
84         new Balen( pin: "");
85     }
86
87
88
89 }
90
```

Fastc.java



```
1 package bank.management.system;
2
3 import javax.swing.*;
4 import java.awt.*;
5 import java.awt.event.ActionEvent;
6 import java.awt.event.ActionListener;
7 import java.sql.ResultSet;
8 import java.util.Date;
9
10 public class Fastc extends JFrame implements ActionListener {
11
12     JButton b1,b2,b3,b4,b5,b6,b7; 7 usages
13
14     String pin; 5 usages
15
16     Fastc(String pin){ 2 usages
17         this.pin = pin;
18
19
20
21         ImageIcon i1 = new ImageIcon(ClassLoader.getResource( name: "icons/atm.jpg"));
22         Image i2 = i1.getImage().getScaledInstance( width: 1550, height: 830,Image.SCALE_DEFAULT);
23         ImageIcon i3 = new ImageIcon(i2);
24         JLabel l3 = new JLabel(i3);
25         l3.setBounds( x: 0, y: 0, width: 1550, height: 830);
26         add(l3);
27
28         JLabel label = new JLabel( text: "SELECT WITHDRAWAL AMOUNT");
29         label.setBounds( x: 510, y: 410, width: 400, height: 35);
30         label.setForeground(Color.BLACK);
31         label.setFont(new Font( name: "System",Font.ITALIC, size: 20));
32         l3.add(label);
33     }
34
35     public static void main(String[] args) {
36         new Fastc( pin: "");
37     }
38 }
39
```

**



```
Bank Management System  Version control  Login  2 1 8  X

Project  Wit.java  Fastc.java  Signup2.java  Signup3.java  Signup.java  Deposit.java  Pinc.java  Balen.java  Login.java  X
Bank Management System
  .idea
  out
  src
    bank.managemen
      Balen
      Co
      Deposit
      Fastc
      Login
      main_Class
      Minis
      Pinc
      Signup
      Signup2
      Signup3
      Wit
    icons
  .gitignore
  Bank Management S
External Libraries
  < 23 > C:\Program F
  jcalendar-tz-1.3.3-4
  mysql-connector-jav
Scratches and Console:

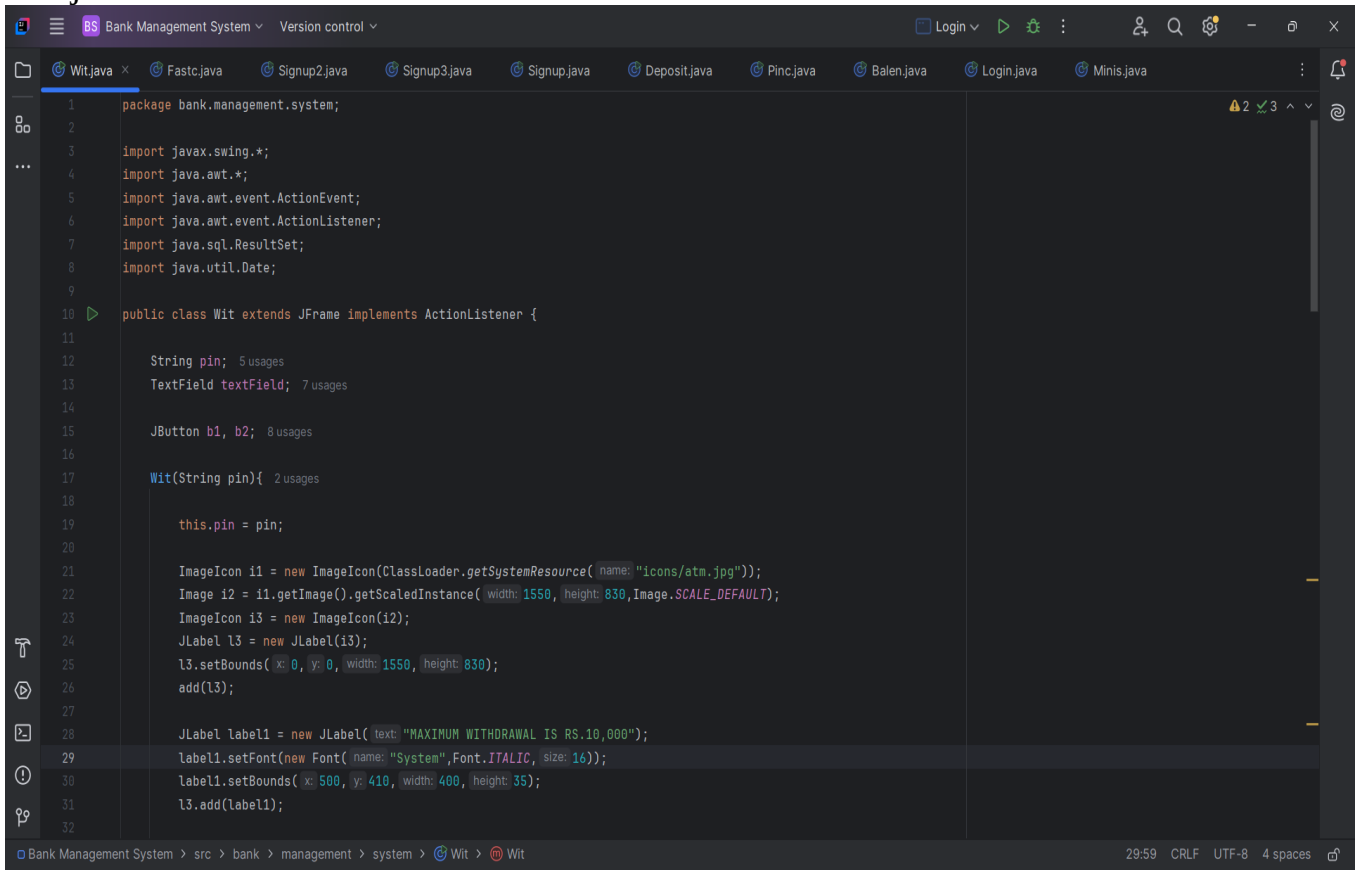
10  public class Fastc extends JFrame implements ActionListener {
17  Fastc(String pin){ 2 usages
93      setLocation(0, 0);
94      setVisible(true);
95
96
97
98
99
100 }
101
102
103 @Override
104 public void actionPerformed(ActionEvent e) {
105     if(e.getSource() == b7){
106         setVisible(false);
107         new main_Class(pin);
108
109     } else{
110         String amount = ((JButton)e.getSource()).getText().substring(4);
111         Co c = new Co();
112         Date date = new Date();
113         try{
114             ResultSet resultSet = c.statement.executeQuery("select * from bank where pin = '"+pin+"'");
115             int balance = 0;
116             while(resultSet.next()){
117                 if(resultSet.getString("columnLabel: type").equals("Deposit")){
118                     balance += Integer.parseInt(resultSet.getString("columnLabel: amount"));
119                 }else{
120                     balance -= Integer.parseInt(resultSet.getString("columnLabel: amount"));
121                 }
122             }String num = "17";
123
124
125
126
127
128
129
130
131
132
133
134
135
136
137
138
139
140
141
142
143
144
145
146
147
```

```
Bank Management System  Version control  Login  2 1 8  X

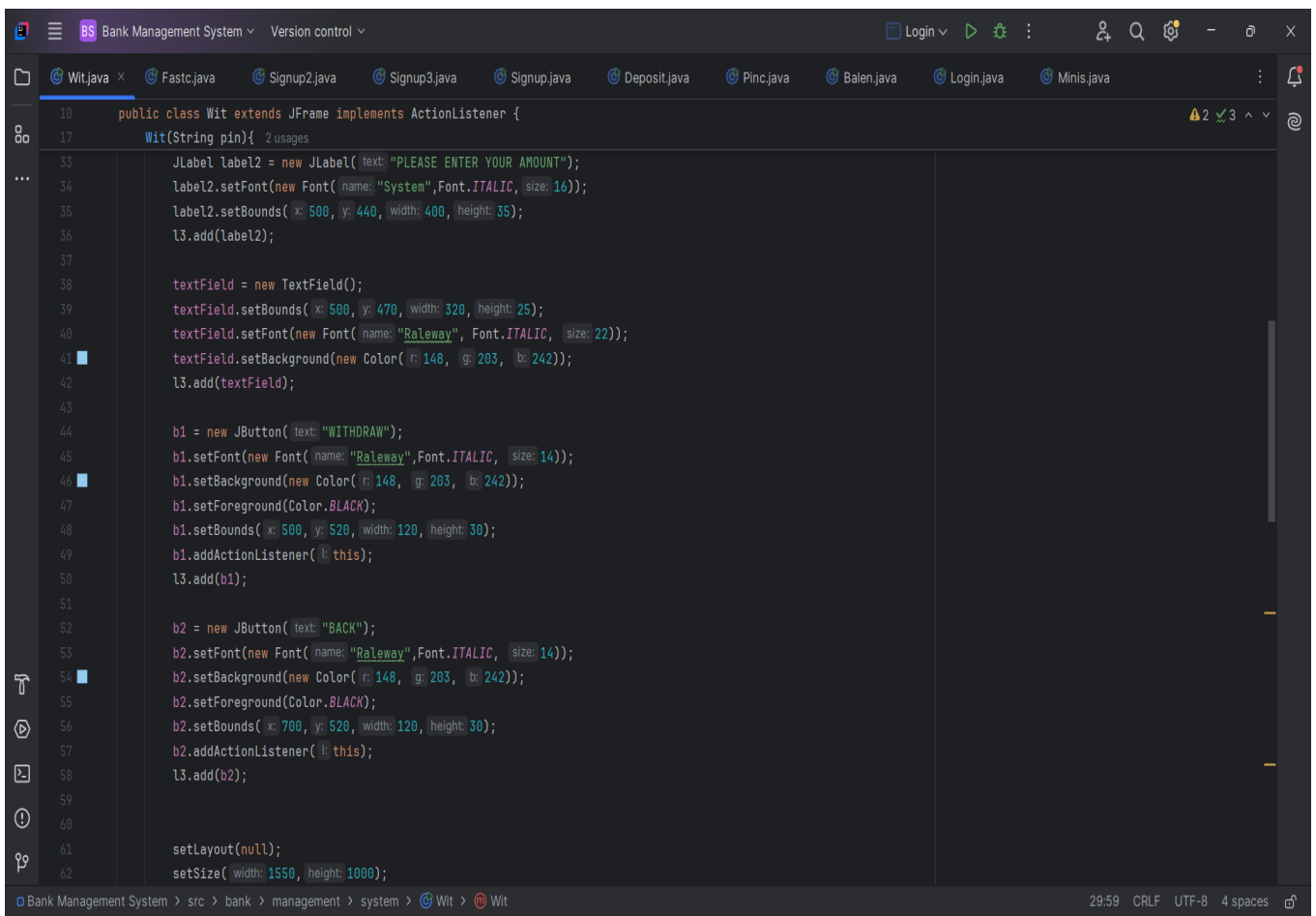
Project  Wit.java  Fastc.java  Signup2.java  Signup3.java  Signup.java  Deposit.java  Pinc.java  Balen.java  Login.java  X
Bank Management System
  .idea
  out
  src
    bank.managemen
      Balen
      Co
      Deposit
      Fastc
      Login
      main_Class
      Minis
      Pinc
      Signup
      Signup2
      Signup3
      Wit
    icons
  .gitignore
  Bank Management S
External Libraries
  < 23 > C:\Program F
  jcalendar-tz-1.3.3-4
  mysql-connector-jav
Scratches and Console:

10  public class Fastc extends JFrame implements ActionListener {
104  public void actionPerformed(ActionEvent e) {
123
124      if(e.getSource() != b7 && balance < Integer.parseInt(amount)){
125          JOptionPane.showMessageDialog(parentComponent: null, message: "Insufficient Balance");
126          return;
127      }
128
129      c.statement.executeUpdate("insert into bank values('"+pin+"', '"+date+"', 'withdrawal', '"+amount+"')");
130      JOptionPane.showMessageDialog(parentComponent: null, message: "Rs. "+amount+"Debited Successfully");
131
132      }catch(Exception E){
133          E.printStackTrace();
134      }
135      setVisible(false);
136      new main_Class(pin);
137  }
138
139
140
141
142
143  public static void main(String[] args) {
144      new Fastc("pin: ");
145  }
146
147
```

Wit.java



```
1 package bank.management.system;
2
3 import javax.swing.*;
4 import java.awt.*;
5 import java.awt.event.ActionEvent;
6 import java.awt.event.ActionListener;
7 import java.sql.ResultSet;
8 import java.util.Date;
9
10 public class Wit extends JFrame implements ActionListener {
11
12     String pin; 5 usages
13     TextField textField; 7 usages
14
15     JButton b1, b2; 8 usages
16
17     Wit(String pin){ 2 usages
18
19         this.pin = pin;
20
21         ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource("icons/atm.jpg"));
22         Image i2 = i1.getImage().getScaledInstance(1550, 830, Image.SCALE_DEFAULT);
23         ImageIcon i3 = new ImageIcon(i2);
24         JLabel l3 = new JLabel(i3);
25         l3.setBounds(0, 0, 1550, 830);
26         add(l3);
27
28         JLabel label1 = new JLabel(text: "MAXIMUM WITHDRAWAL IS RS.10,000");
29         label1.setFont(new Font(name: "System", Font.ITALIC, size: 16));
30         label1.setBounds(x: 500, y: 410, width: 400, height: 35);
31         l3.add(label1);
32
```



```
33         JLabel label2 = new JLabel(text: "PLEASE ENTER YOUR AMOUNT");
34         label2.setFont(new Font(name: "System", Font.ITALIC, size: 16));
35         label2.setBounds(x: 500, y: 440, width: 400, height: 35);
36         l3.add(label2);
37
38         textField = new TextField();
39         textField.setBounds(x: 500, y: 470, width: 320, height: 25);
40         textField.setFont(new Font(name: "Raleway", Font.ITALIC, size: 22));
41         textField.setBackground(new Color(r: 148, g: 203, b: 242));
42         l3.add(textField);
43
44         b1 = new JButton(text: "WITHDRAW");
45         b1.setFont(new Font(name: "Raleway", Font.ITALIC, size: 14));
46         b1.setBackground(new Color(r: 148, g: 203, b: 242));
47         b1.setForeground(Color.BLACK);
48         b1.setBounds(x: 500, y: 520, width: 120, height: 30);
49         b1.addActionListener(b: this);
50         l3.add(b1);
51
52         b2 = new JButton(text: "BACK");
53         b2.setFont(new Font(name: "Raleway", Font.ITALIC, size: 14));
54         b2.setBackground(new Color(r: 148, g: 203, b: 242));
55         b2.setForeground(Color.BLACK);
56         b2.setBounds(x: 700, y: 520, width: 120, height: 30);
57         b2.addActionListener(b: this);
58         l3.add(b2);
59
60         setLayout(null);
61         setSize(width: 1550, height: 1000);
62
```

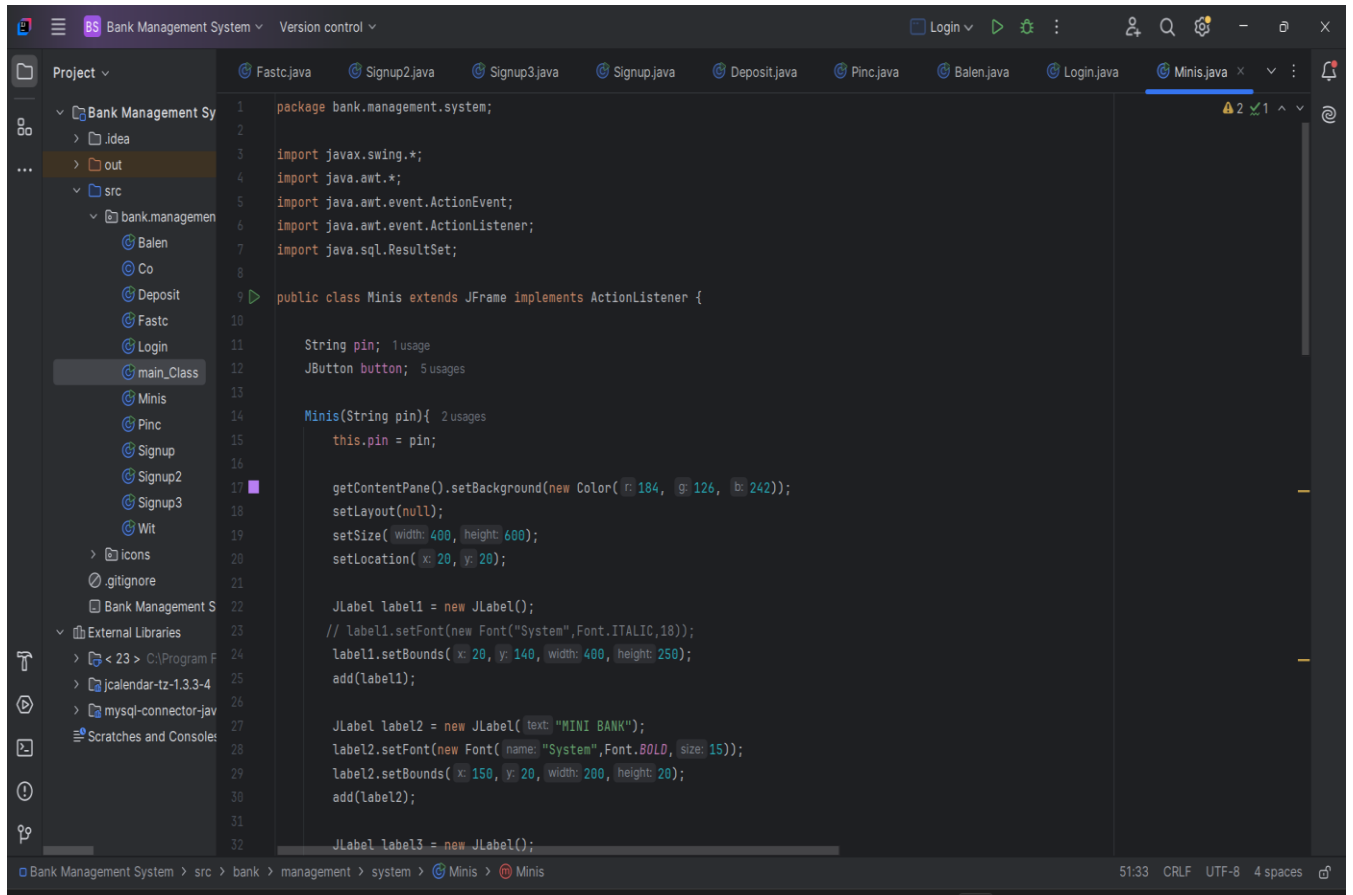
**

```
Bank Management System Version control Login Run Debug Search View Help
Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java
10 public class Wit extends JFrame implements ActionListener {
17 Wit(String pin){ 2 usages
63 setLocation( x: 0, y: 0);
64 setVisible(true);
65
66
67
68
69
70
71
72 @Override
73 public void actionPerformed(ActionEvent e) {
74     if(e.getSource() == b1) {
75         try {
76             String amount = textField.getText();
77             Date date = new Date();
78             if (textField.getText().equals("")) {
79                 JOptionPane.showMessageDialog( parentComponent: null, message: "Please enter the amount you want to withdraw");
80             } else {
81                 Co c = new Co();
82                 ResultSet resultSet = c.statement.executeQuery( sql: "select * from bank where pin = '"+pin+"'");
83                 int balance = 0;
84                 while (resultSet.next()) {
85                     if (resultSet.getString( columnLabel: "type").equals("Deposit")){
86                         balance += Integer.parseInt(resultSet.getString( columnLabel: "amount"));
87                     } else {
88                         balance -= Integer.parseInt(resultSet.getString( columnLabel: "amount"));
89                     }
90                 }
91                 if (balance < Integer.parseInt(amount)) {
92                     JOptionPane.showMessageDialog( parentComponent: null, message: "Insufficient Balance");
93                 }
94             }
95         } catch (Exception e) {
96             JOptionPane.showMessageDialog( parentComponent: null, message: "Insufficient Balance");
97         }
98     } else if(e.getSource() == b2){
99         setVisible(false);
100         new main_Class(pin);
101     }
102 }
103
104 public static void main(String[] args) { new Wit( pin: ""); }
105 }
106
Bank Management System > src > bank > management > system > Wit > Wit 29:59 CRLF UTF-8 4 spaces
```

```
Bank Management System Version control Login Run Debug Search View Help
Wit.java Fastc.java Signup2.java Signup3.java Signup.java Deposit.java Pinc.java Balen.java Login.java Minis.java
10 public class Wit extends JFrame implements ActionListener {
72 public void actionPerformed(ActionEvent e) {
90     if (balance < Integer.parseInt(amount)) {
91         JOptionPane.showMessageDialog( parentComponent: null, message: "Insufficient Balance");
92         return;
93     }
94
95     c.statement.executeUpdate( sql: "insert into bank values('" + pin + "', '" + date + "', 'withdrawal', '" + amount + "')");
96     JOptionPane.showMessageDialog( parentComponent: null, message: "Rs. " + amount + "Debited Successfully");
97     setVisible(false);
98     new main_Class(pin);
99 }
100
101 } catch (Exception E) {
102
103 }
104 }else if(e.getSource() == b2){
105     setVisible(false);
106     new main_Class(pin);
107 }
108 }
109
110
111
112 public static void main(String[] args) { new Wit( pin: ""); }
113 }
114
115
116
Bank Management System > src > bank > management > system > Wit > Wit 29:59 CRLF UTF-8 4 spaces
```

**

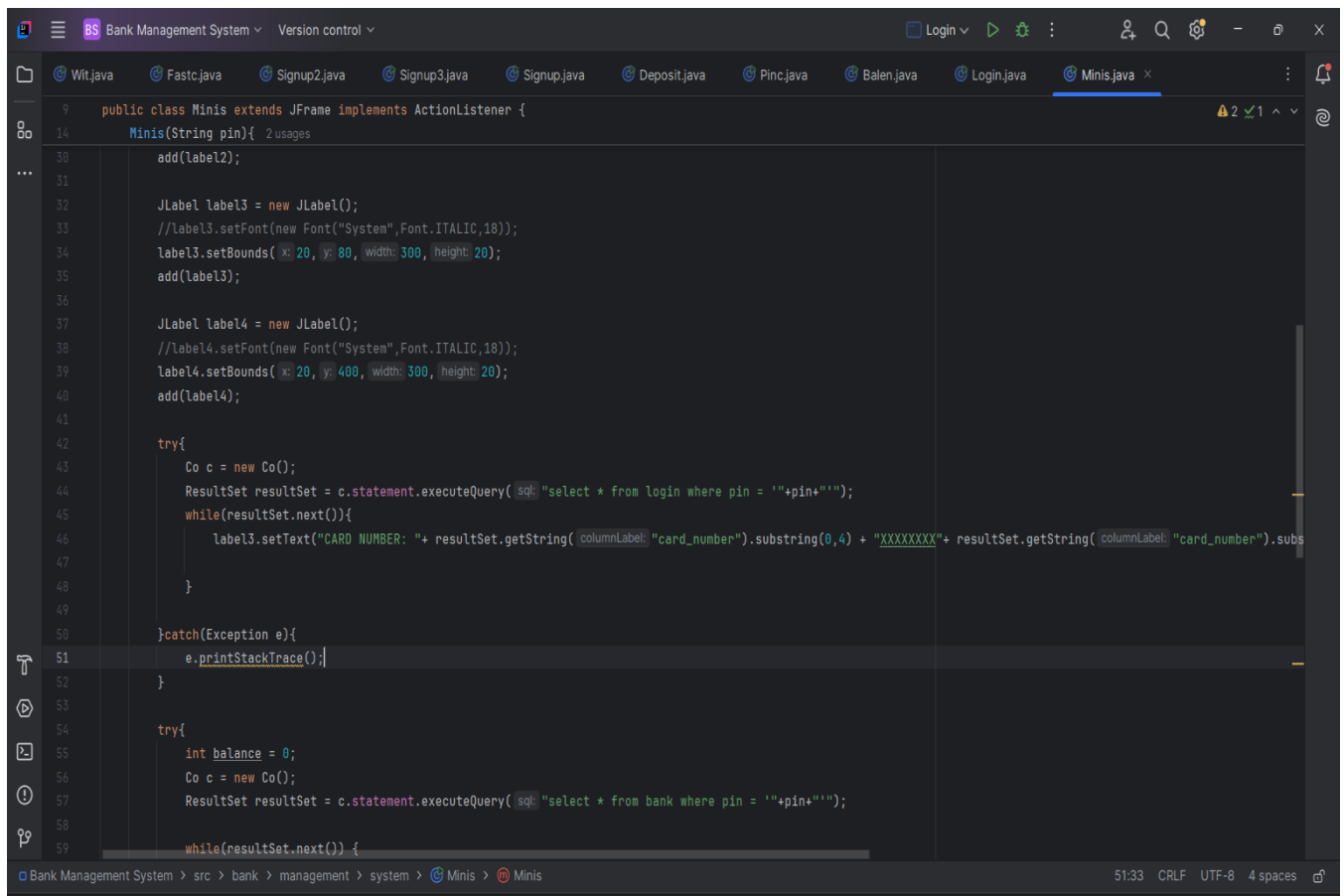
Minis.java



The screenshot shows an IDE window titled 'Bank Management System' with a 'Version control' dropdown. The project structure on the left includes 'Bank Management System' with subfolders 'out' and 'src'. The 'src' folder contains a 'bank.management' package with classes 'Balen', 'Co', 'Deposit', 'Fastc', 'Login', 'main_Class', 'Minis', 'Pinc', 'Signup', 'Signup2', 'Signup3', and 'Wit'. The 'Minis.java' file is open in the editor, showing the following code:

```
1 package bank.management.system;
2
3 import javax.swing.*;
4 import java.awt.*;
5 import java.awt.event.ActionEvent;
6 import java.awt.event.ActionListener;
7 import java.sql.ResultSet;
8
9 public class Minis extends JFrame implements ActionListener {
10
11     String pin; 1 usage
12     JButton button; 5 usages
13
14     Minis(String pin){ 2 usages
15         this.pin = pin;
16
17         getContentPane().setBackground(new Color(184, 126, 242));
18         setLayout(null);
19         setSize(400, 600);
20         setLocation(20, 20);
21
22         JLabel label1 = new JLabel();
23         // label1.setFont(new Font("System",Font.ITALIC,18));
24         label1.setBounds(20, 140, 400, 250);
25         add(label1);
26
27         JLabel label2 = new JLabel("MINI BANK");
28         label2.setFont(new Font("System",Font.BOLD, 15));
29         label2.setBounds(150, 20, 200, 20);
30         add(label2);
31
32         JLabel label3 = new JLabel();
```

The status bar at the bottom indicates 'Bank Management System > src > bank > management > system > Minis > Minis' and '51:33 CRLF UTF-8 4 spaces'.



The screenshot shows the continuation of the 'Minis.java' file in the IDE. The code continues from the previous snippet:

```
33         //label3.setFont(new Font("System",Font.ITALIC,18));
34         label3.setBounds(20, 80, 300, 20);
35         add(label3);
36
37         JLabel label4 = new JLabel();
38         //label4.setFont(new Font("System",Font.ITALIC,18));
39         label4.setBounds(20, 400, 300, 20);
40         add(label4);
41
42         try{
43             Co c = new Co();
44             ResultSet resultSet = c.statement.executeQuery("select * from login where pin = '"+pin+"'");
45             while(resultSet.next()){
46                 label3.setText("CARD NUMBER: " + resultSet.getString("card_number").substring(0,4) + "XXXXXX" + resultSet.getString("card_number").subs
47             }
48         }
49         catch(Exception e){
50             e.printStackTrace();
51         }
52     }
53
54     try{
55         int balance = 0;
56         Co c = new Co();
57         ResultSet resultSet = c.statement.executeQuery("select * from bank where pin = '"+pin+"'");
58         while(resultSet.next()) {
```

The status bar at the bottom indicates 'Bank Management System > src > bank > management > system > Minis > Minis' and '51:33 CRLF UTF-8 4 spaces'.

**

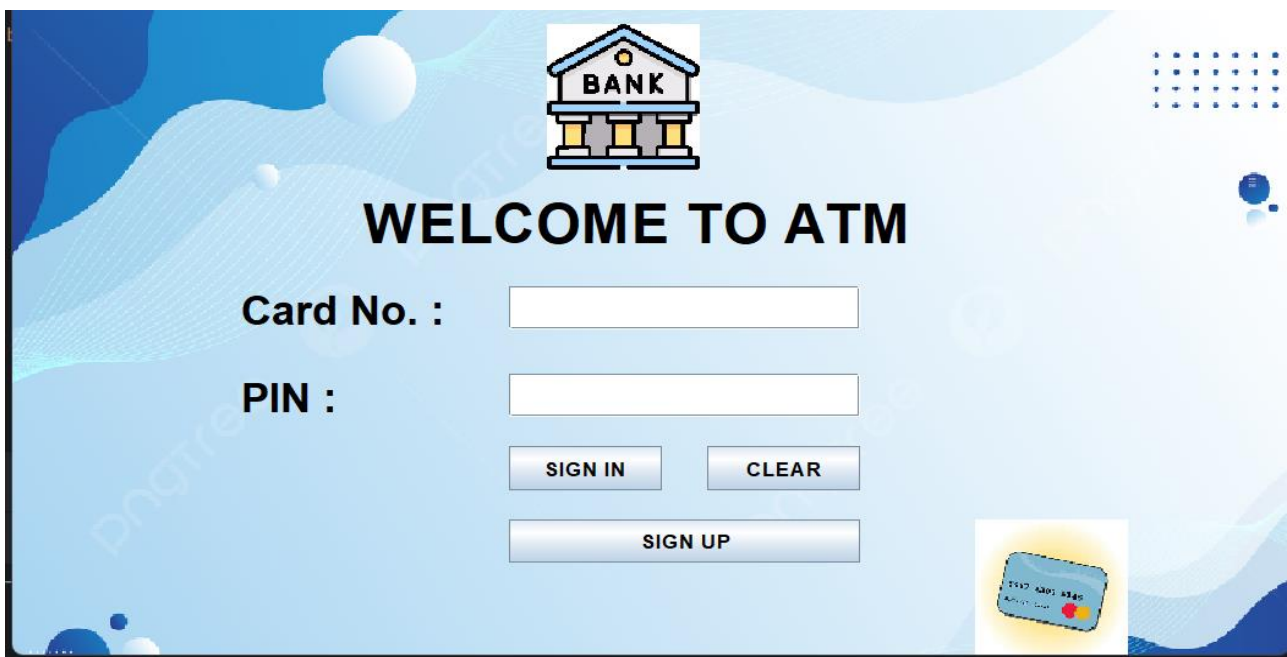
My SQL

4.4 DATABASE BANK SYSTEM

- create database bankSystem;
- use bankSystem;
- create table signup(form_no varchar(30), name varchar(30), father_name varchar(30), DOB varchar(30), gender varchar(30), email varchar(60), marital_status varchar(30), address varchar(60), city varchar(30), pincode varchar(30), state varchar(50));
- select * from signup;
- create table signuptwo(form_no varchar(30), religion varchar(30), category varchar(30), income varchar(30), education varchar(30), occupation varchar(60), pan varchar(30), aadhar varchar(60), seniorcitizen varchar(30), existing_account varchar(30));
- select * from signuptwo;
- create table signupthree(form_no varchar(30), account_type varchar(40), card_number varchar(30), pin varchar(30), facility varchar(200));
- select * from signupthree;
- create table login(form_no varchar(30), card_number varchar(50), pin varchar(30));
- select * from login;
- create table bank(pin varchar(10), date varchar(50), type varchar(20), amount varchar(20));
- select * from bank;

4.5 OUTPUT OF THE MODEL

Login.java

The image shows a graphical user interface for an ATM system. At the top center is a logo of a bank building with the word "BANK" on it. Below the logo, the text "WELCOME TO ATM" is displayed in a large, bold, black font. Underneath this, there are two input fields: "Card No. :" followed by a white rectangular box, and "PIN :" followed by another white rectangular box. Below the PIN field, there are three buttons: "SIGN IN" and "CLEAR" are side-by-side, and "SIGN UP" is centered below them. In the bottom right corner, there is a small image of a credit card. The background is a light blue gradient with some abstract shapes and a grid of small dots in the top right corner.

BANK

WELCOME TO ATM

Card No. :

PIN :

SIGN IN **CLEAR**

SIGN UP



**

Signup.java



APPLICATION FORM NO. 3487

Page 1
Personal Details

NAME :

FATHER'S NAME :

DATE OF BIRTH :

GENDER :

☐ MALE

☐ FEMALE

EMAIL ADDRESS :

MARITAL STATUS :

☐ MARRIED

☐ UNMARRIED

☐ OTHER

ADDRESS :

CITY :

PINCODE :

STATE :

NEXT

Signup2.java



FORM No. :

Page 2
Additional Details

RELIGION :

Hinduism

CATEGORY :

General

INCOME :

Null

EDUCATIONAL :

Non-Graduate

OCCUPATION :

Salaried

AADHAR NUMBER :

PAN NUMBER :

SENIOR CITIZEN :

☐ YES

☐ NO

EXISTING ACCOUNT :

☐ YES

☐ NO

NEXT

**

Signup3.java



FORM No. :
Page 3
Account Details

ACCOUNT TYPE :

☐ SAVING ACCOUNT

☐ CURRENT ACCOUNT

☐ FIXED DEPOSIT ACCOUNT

☐ RECURRING DEPOSIT ACCOUNT

CARD NUMBER :
(Your 16-Digit Card Number)

XXXX-XXXX-XXXX-4841
(It would appear on atm card/cheque book and statements)

PIN :
(4-digit Password)

XXXX

SERVICES REQUIRED :

☐ ATM CARD

☐ INTERNET BANKING

☐ MOBILE BANKING

☐ EMAIL ALERTS

☐ CHEQUE BOOK

☐ E-STATEMENT

☒ I here by declares that the above entered details correct to the best of my knowledge.

SUBMIT

CANCEL

After submitting the all three forms a pop up message will be generate like shown below



FORM No. : 1463
Page 3
Account Details

ACCOUNT TYPE :

☐ SAVING ACCOUNT

☐ CURRENT ACCOUNT

☒ FIXED DEPOSIT ACCOUNT

☐ RECURRING DEPOSIT ACCOUNT

CARD NUMBER :
(Your 16-Digit Card Number)

XXXX-XXXX-XXXX-4841
(It would appear on atm card/cheque book and statements)

PIN :
(4-digit Password)

SERVICES REQUIRED :

☒ ATM CARD

☐ INTERNET BANKING

☐ MOBILE BANKING

☐ EMAIL ALERTS

☐ CHEQUE BOOK

☐ E-STATEMENT

☒ I here by declares that the above entered details correct to the best of my knowledge.

SUBMIT

CANCEL

Message
CARD NUMBER :1409962989928489
PIN :4062
OK

**

Access to ATM and its Facilities



Deposit.java



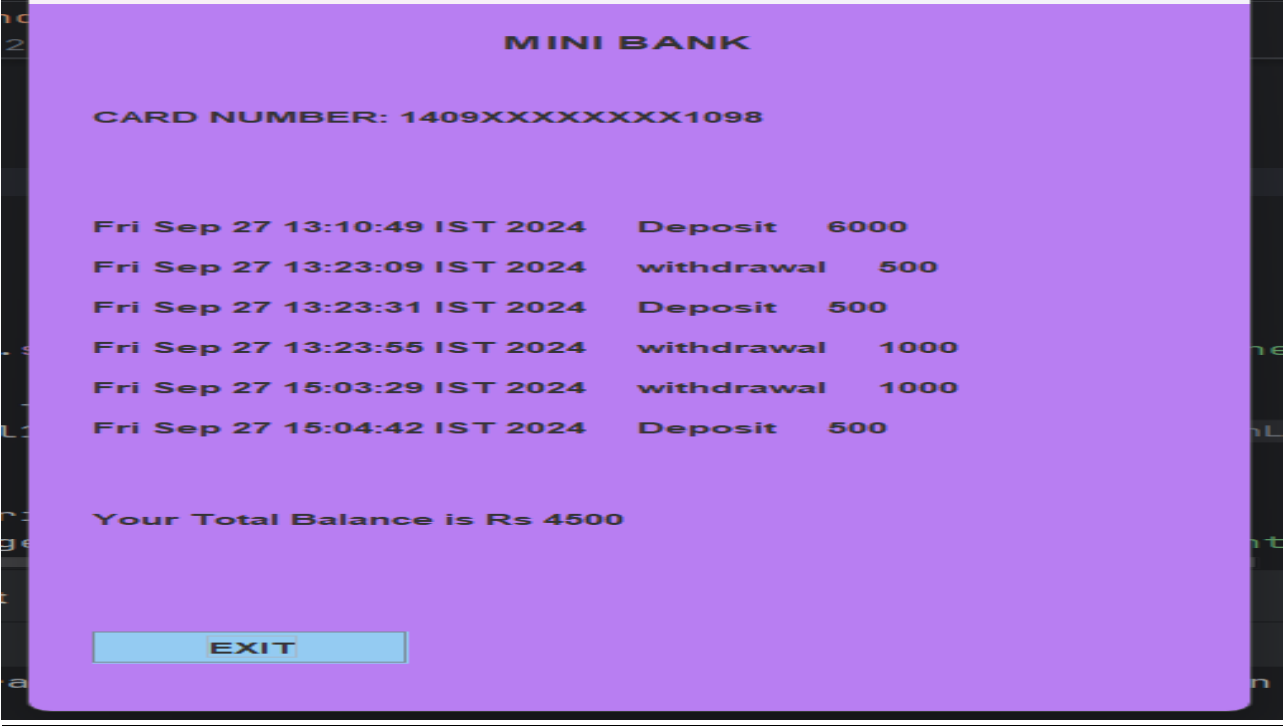
Wit.java



Fastc.java



Minis.java



Pin.java



Balan.java



CHAPTER 5

5 CONCLUSION

5.1 Scope

- Creating Accounts- The program can be used to create accounts by the customers. It helps save the hustle for the customer to visit the bank physically and create/use these accounts.
- Depositing Money
- Withdrawing Money
- Balance Enquiry- The customer can check their balance via this program.
- Changing Passwords/PIN- The customer can easily change the passwords and pin numbers using the program.
- Mini Statement
- Fast Cash facilities
- Cardless ATM withdrawals

5.2 pros and cons

Pros of using cardless ATMs

1. Fewer steps involved

“Our phone is slowly becoming our wallet,” says Drew Cheneler, founder of SimpleMoneyLyfe. Some consumers don’t memorize their account information and rely on facial or fingerprint recognition to access their accounts, he says. Some cardless ATM software allows you to do the same.

2. Access to all your accounts

You might not carry all your ATM cards with you, especially if you use more than one bank. Or, you might have forgotten or lost your card. Cardless ATMs allow you to still have access to all of your accounts, which can be helpful if you are near an in-network ATM, yet don’t have the corresponding card with you, saving you from having to pay out-of-network ATM fees.

3. No need to carry a wallet

It isn’t always convenient to carry a wallet, purse or something else containing your cards. Cardless ATMs allow you to perform transactions if you have your smartphone. Another benefit of not carrying your wallet is it can reduce your chances of getting pickpocketed or robbed.

4. Security

Though they aren’t risk-free, a few features of cardless ATM withdrawals make them more secure. For starters, scammers can’t use skimmers to steal your card data because you don’t insert your card into the ATM terminal. Codes generated for cardless withdrawals are one-time use only, so it does a hacker little good to steal one. Cardless withdrawals also often require two-step authentication, so hackers wouldn’t only need to know your PIN, but they would also need to have access to your phone.

5. A more sanitary option

Cardless ATM withdrawals won’t eliminate the need to touch the ATM, yet they do cut down on contact. And the fewer touches the better, considering how many users might touch an ATM and spread germs between cleanings.

Cons of using cardless ATMs

1. Availability

Cardless ATMs aren't available everywhere. Even if your bank provides the service and your phone is compatible with the technology, you might not find an ATM near you that's capable of handling cardless withdrawals.

2. Compatibility

Cardless ATMs rely on smartphones to operate. If your bank utilizes an app, your phone must be compatible with that app, or you can't use it. Furthermore, not all smartphones are compatible with cardless ATMs.

3. Phone security

Though cardless ATM transactions are generally more secure, they pose a different set of risks.

"Since bank cards are being replaced by phones, criminals will now be more enticed to target and hack your phone to mine your data, information and even change your phone and bank app settings," says Ricardo Pina, founder of The Modest Wallet. You can help to minimize the risk by keeping your phone secure if you have banking apps installed.

Popular accounts with cardless ATM access

Chase, Wells Fargo, Capital One and Bank of America are all compatible with Google Pay, Apple Pay and Samsung Pay. Since cardless ATM apps from these banks work on each of these services, they are compatible with both Android and Apple phones.

Bottom line

Cardless ATMs are a convenient, secure and more sanitary way to withdraw cash without a card. They're also handy for emergencies — even if you don't use them regularly. Look out for the next generation of cardless banking to be even more high-tech, too, with the potential to leave your card and your phone at home.

5.3 Applications

Bank Management System is a Java program that allows users to manage their bank accounts. It is built using the Java programming language, the Swing graphical user interface library, and the MySQL database management system.

And can be introduced as a cardless atm services.

5.4 Limitations

- As INDIA is forwarding toward cashless country through Make in India initiative ,it is may be not that usefull in upcoming time.
- Security is also an issue here .
- Introduction of cardless ATM in our country may not be that advantage due to production cost and security issue .

Although INDIA is the most populated country in the world , cardless ATM will be the better option due to sanitary reasons. As pandemic taught us about the cleanliness many country from that time to till now are using this kind of service.

REFERENCES

--Google Chrome

-GeeksforGeeks

-Coding Ninja

-<https://www.bankrate.com/banking/how-do-cardless-atms-work/>

--Youtube

-TechcoderAV

-Apna college

-Chat GPT

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